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Review article

A reappraisal of the deconditioning hypothesis in low back pain: review of evidence from a triumvirate of research methods on specific lumbar extensor deconditioning

James Steele
Stewart Bruce-Low
Southampton Solent University, Southampton, Hampshire, UK

Dave Smith
Manchester Metropolitan University, Crewe, Cheshire, UK

Address for correspondence:
Centre for Health, Exercise and Sport Science, Southampton Solent University, East Park Terrace, Southampton, Hampshire SO14 0YN, UK.
Tel: +442380 319606; Fax: +442380 332627; james.steele@solent.ac.uk

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Abstract

Objective:
'Disuse' and 'Deconditioning' in relation to low back pain (LBP) are terms often used interchangeably. Discussions of 'disuse' refer to general physical inactivity, which evidence suggests does not differ between symptomatic and asymptomatic persons. 'Deconditioning' refers to a decrease in function, commonly both cardiovascular/aerobic fitness and muscular strength/endurance, again noting little difference. However, examination of decreased function relating specifically to lumbar extensor musculature deconditioning has yet to be examined, corroborating all possible methods. Thus, this review attempts to reappraise the deconditioning hypothesis in LBP, specifically considering lumbar extensor deconditioning.

Methods:
A literature review was conducted examining both cross-sectional and prospective data on specific lumbar extensor deconditioning and LBP. A narrative approach and 'snowballing' style literature search was used involving initial use of PubMed and Google Scholar databases searching up to December 2012. Included were studies utilizing the following three research methods, allowing specific induction of the role of such deconditioning; (1) strength/endurance testing of the isolated lumbar extensor musculature, (2) imaging and histochemical examination of the lumbar extensor musculature, and (3) fatigue testing of the lumbar extensor musculature using electromyography.

Results/findings:
Despite issues interpreting individual studies due to methods, the majority of evidence suggests LBP is associated with decreased strength/endurance, atrophy, and excessive fatigability of the lumbar extensors. Prospective studies also suggest lumbar extensor deconditioning may be a common risk factor predicting acute low back injury and LBP.

Conclusions:
The hypothesis of specific lumbar extensor deconditioning as being a causal factor in LBP is presently well supported. It is by no means the only causative factor and further research should more rigorously test this hypothesis addressing the methodological issues highlighted regarding previous studies. However, its role suggests specific exercise may be a worthwhile preventative and rehabilitative approach.

Introduction

Defining the disuse/deconditioning hypothesis—'disuse' or 'deconditioning'?

The 'Disuse Syndrome' was originally described by Bortz II[1] and more recently has been reviewed by Verbunt *et al.*[2,3]. The rationale behind Disuse Syndrome is that pain causes low levels of physical activity (i.e., avoidance behavior or guarded movement[4]) which contribute to deconditioning and chronicity in low back pain (LBP), and cause the further inter-related physical and psychological changes shown in Figure 1. In essence, it proposes that injury and pain precede deconditioning and potentially many of LBP's symptoms, leading to a 'vicious cycle.'

Verbunt *et al.*[2] however, have suggested that the hypothesis that 'disuse' (i.e., defined as a *decrease in physical activity levels*) is a cause of LBP may be incorrect. They highlight that activity levels are in fact similar between symptomatic and asymptomatic participants, suggesting that lack of physical activity due to the presence of pain or injury may not contribute to the presence of deconditioning or LBP[5]. Indeed a more recent study has also highlighted that physical activity levels appear to not change as a result of LBP, even as it develops from acute into chronic LBP (CLBP)[6]. This suggests that symptomatic CLBP participants may not suffer from development of disuse after the initial incidence of LBP. It seems possible, therefore, that the direction of temporal relationships in the 'Disuse Syndrome' model may be unfitting in its usual presentation (Figure 1). This is not to suggest that deconditioning as a result of existing pain and its related behaviors is not a possibility, indeed the presence of injury has been shown to affect muscular function and could, therefore, instigate deconditioning itself, or at the very least further enhance its development[7,8].

Instead, 'deconditioning' (i.e., defined as a *decrease in function*) may be first implicated as a potential cause of low back injury and pain, as opposed to LBP leading to 'disuse' and then to 'deconditioning'. Indeed, Verbunt *et al.*[2] attempt to distinguish between 'disuse' and 'deconditioning'; however, in both their definitions they inevitably invoke general physical inactivity ('disuse') as being responsible for 'deconditioning' and that this inactivity is the result of pain. Here we instead differentiate between 'disuse' and 'deconditioning' and pose that the disuse syndrome model does not consider what first causes or increases the probability of injury or pain occurring (which *may* lead to further 'disuse') in the first place.

In addition, the disuse model appears to imply that freak injury may account for the majority of LBP[7], yet due to the widespread prevalence of LBP it seems unlikely that freak injury could account for the majority of cases. Bigos *et al.*[9] demonstrate exactly why this is a concern, reporting that accidents such as slips or falls, despite resulting in higher cost injuries, are very uncommon with regard to cause of injury; lifting or materials handling, however, was most commonly considered a cause. Dysfunction due to deconditioning could potentially affect such actions, leading to

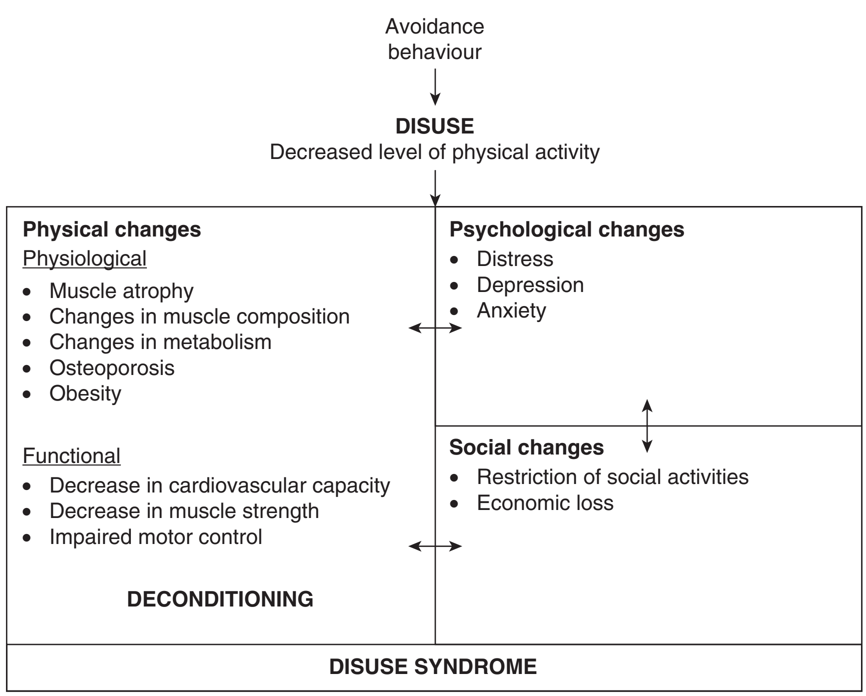

Figure 1. Disuse syndrome model, from Verbunt *et al.*[2].

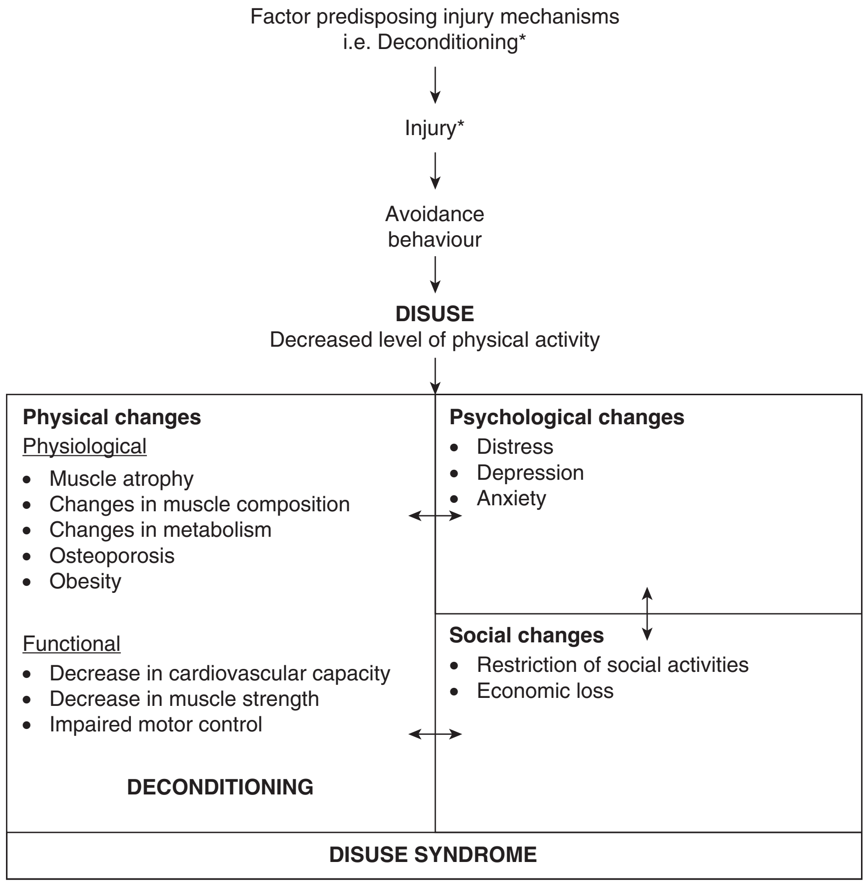

Figure 2. Deconditioning syndrome model–adapted * Disuse syndrome model–adapted from Verbunt *et al.*[2].

fatigue and altered joint biomechanics, subsequently causing injuries and instigating mechanisms by which pain results. Then, at this point, the cycle by which a reduction in activity levels further promote chronicity, and the changes associated with it, may begin to have an influence.

The model may, therefore, require an addition that considers the initial injury in the first place (Figure 2). A high percentage of low back injury and acute LBP develops into CLBP[10,11] and so it seems logical that something must affect the risk of low back injury in the first instance. Indeed, Adams *et al.*[12] have recently commented on the pertinent fact that all 'chronic back pain always starts as acute back pain'. Thus, logically, something must first be responsible for the initiation of acute pain. The remainder of the existing model is likely correct in describing the process of developing chronicity after initial injury has occurred. What specific factor is most important in determining whether that initial injury and acute pain occurs in order for it to become chronic, however, is the more interesting question.

Many prior reviews on the topic of deconditioning in LBP have utilized a broad focus encompassing decrease in function of both cardiovascular/aerobic fitness as well as muscular strength/endurance[2,3,13,14]. These reviews suggest that deconditioning of these kinds is not apparent in those with CLBP. However, the studies considered have utilized varied methods of examining this association, many of which are not entirely specific, and these are explained throughout this article. The lack of consideration of the specific study methodologies used by previous authors has perhaps contributed to the vague distinctions between 'disuse' and 'deconditioning' as well as the shift in focus from physical risk factors towards a more cognitive based appraisal of LBP and the effects of rehabilitation[15]. A more specific analysis of the literature, focusing upon specifically located deconditioning and in consideration of the methodological limitations of prior techniques, might, therefore, be found to yield contrasting conclusions regarding the presence of deconditioning in LBP and the models relationships.

Specific disuse/deconditioning of the lumbar extensors

The important role of the lumbar extensor musculature, the Erector Spinae (ES; i.e., iliocostalis lumborum and longissimus thoracis) and both deep and superficial lumbar Multifidus (MF), in providing stability to the lumbar spine, has been alluded to in numerous studies[16–24]. Prior reviews of the literature have indicated that, although it is difficult to distinguish which muscles provide the greatest relative contribution to spinal stability, their importance in co-operatively contributing to lumbar spinal stability is clear[23,25]. The relative contribution of individual muscles will vary depending on the specific task being performed[26]; however, MacDonald *et al.*[23] explain that all lumbar extensors can contribute towards stability of the intervertebral segments through compression of the vertebral unit and increased joint stiffness. Clearly the lumbar extensor musculature playing such an important role in providing stability to the lumbar spine suggests that deconditioning and dysfunction in these muscles could lead to changes in stability and biomechanics. This change in biomechanics may result in increasing passive tissue stresses and potentially impart an injury or pain response in structures of the lumbar spine[27].

Aim and approach of this review

The focus upon 'disuse' and 'deconditioning' in a general sense has led to much incongruity in drawing specific conclusions regarding LBP. In light of the potential role of the lumbar extensors in controlling stability and, in dysfunction, altering biomechanics which might lead to injury and pain causing mechanisms/symptoms, there is certainly potential for specifically located deconditioning to relate to LBP. The aim of this review, therefore, is to test this hypothesis by examining the evidence reporting the nature of the relationship between specific deconditioning of the lumbar extensors and LBP from a triumvirate of research methods including:

- Strength/endurance testing of lumbar extensor musculature,
- Imaging and histochemical examination of the lumbar extensor musculature, and
- Fatigue testing of the lumbar extensor musculature using electromyography.

Three sections will follow, each covering the three research methods highlighted as they have been used in cross-sectional examination of symptomatic CLBP participants compared with asymptomatic healthy participants. A fourth section shall examine prospective studies that have sought to examine the effect of deconditioning using these methods upon development of LBP in asymptomatic participants. Throughout, any methodological concerns and considerations with studies shall be highlighted initially and noted whilst discussing such studies. In light of those methodological issues discussed in each section it will also be noted which types of studies were excluded from consideration (however, those excluded are still summarized within the full summary tables in the Supplementary Appendices). Given the broad scope of this review, a narrative approach utilizing a 'snowballing' style literature search[28] was used initially involving PubMed and Google Scholar databases searching up to December 2012 utilizing search terms including combinations and synonyms of 'low back pain', 'low back injury', 'lumbar', 'back', 'spine', 'extensors', 'lumbar extension', 'trunk extension', 'erector spinae', 'multifidus', 'iliocostalis lumborum', 'longissimus thoracis', 'strength', 'endurance', 'atrophy', 'cross-sectional area', 'fat infiltration', 'muscle density', 'histochemistry', 'fiber type', 'electromyography', 'fatigability', etc. In addition, previous reviews and any located articles reference lists were searched. This was selected as the best way to locate, examine, and synthesize the maximum amount of information in the various sections covered, thus initial inclusion criteria were based upon applicability to this particular area of discussion, and whether studies had utilized the above noted research methods, before applying specific exclusion criteria (noted in each section of this review).

Strength and endurance of the lumbar extensor musculature in LBP

Considerations for studies of strength and endurance of the lumbar extensor musculature

An initial consideration when looking at studies of muscular performance should be that of the false duality between definitions of muscular strength and endurance expressed by many authors, especially within the field of exercise and LBP[29,30]. Muscular endurance can be defined as being *absolute* (i.e., the number of repetitions/time performed at a given resistance) or *relative* (i.e., the number of repetitions/time performed at a given percentage of a one repetition maximum [1RM] or other maximum strength measurement)[31–33]. For example, a pre-training 1RM of 100 kg might produce 10 repetitions at an absolute value of 70 kg, which is also the relative value of 70% 1RM. However, after a training intervention where the 1RM has improved to 120 kg, a participant will almost certainly be capable of greater than 10 repetitions at the absolute value of 70 kg, but likely still only produce a maximum of 10 repetitions at the relative value of 70% 1RM (now 84 kg). This example shows an increase in maximal strength (1RM) leading to an increase in absolute muscular endurance (i.e., an increase in number of repetitions at the fixed submaximal weight). Research supports this concept[34]. However, research does not support the idea that the same is true of relative loads, but rather that similar maximal repetitions are possible[34,35]. In practice, with relevance to the deconditioning hypothesis and for the LBP participant, low strength would translate to low absolute endurance, high strength to high absolute endurance, and vice versa. Therefore, presuming average external loads typically experienced (i.e., in working conditions, etc.) might remain constant, an increase in strength would theoretically mean an increase in endurance at those absolute loads experienced. Thus, it would seem logically erroneous to attempt to draw a distinction between the two and to claim that one is more important than the other with regards to LBP[29]. Indeed, Mannion[7] has commented that the hypothesis of fatigability as being associated with LBP is essentially analogous to the hypothesis of insufficient strength (both being manifestations of lumbar extensor deconditioning relevant to the deconditioning hypothesis).

Numerous studies have sought to identify the relationship between functional measures of strength and endurance of the lumbar extensor muscles. However, it should be highlighted that the validity of a number of methods of tests for strength and/or endurance of the lumbar spine are questionable due to methodological difficulties; a primary concern being whether sufficient pelvic restraints have been utilized. Essentially, tests have either been performed to examine *trunk* extension (TEX) or isolated *lumbar* extension (ILEX; testing utilizing pelvic stabilization through use of a semi-seated position with rear pelvic restraint and a belt across the thighs). In considering lumbar extensor deconditioning TEX studies require careful reflection along with corroboration of more valid test measures, i.e., ILEX. If the pelvis is not stabilized during testing of extension then it is impossible to determine the actual source of measured extension torque during tests of strength and may involve the hip extensors[36–42] contributing to overstate torque measures[43–45], due to the longer moment arms over which the gluteus and hamstrings exert force and their relatively larger cross-sectional areas[46]. At most only 3° of pelvic rotation[47], likely a result of soft tissue compliance, occurs during ILEX testing of this kind. Lack of pelvic restraint perhaps partly explains the inconsistent reproducibility of TEX endurance tests[48–52], as compared with the consistency of ILEX strength and endurance testing[53–56]. Indeed, despite the aforementioned relationship between strength and absolute endurance there is a poor relationship between tests of ILEX strength and TEX endurance[57]. This highlights that the tests may utilize different musculature.

Although tests of ILEX are more valid representations of lumbar extensor function due to TEX being a compound movement requiring additional rotation of the pelvis through the hip extensor musculature[36,37,41–47,58–60], a large number of studies have made use of tests measuring TEX. Smidt *et al.*[43] also explain that consideration of both tests of TEX and ILEX are indeed valuable when interpreted together as they allow both a deductive, and further an inductive, approach to identify the so-called 'weak link' within the kinetic chain and, thus, we initially considered both studies in this review. Beimborn and Morrissey[61] reviewed early literature on trunk muscle performance in LBP suggesting a consistent association with reduced TEX strength in symptomatic participants, as well as further studies. These TEX studies have been summarized in the Supplementary Appendices provided and appear to show inconsistent associations; some results which support the link between TEX strength/endurance and LBP[37,48,62–82], and some which do not[66,69,71,72,81,83–90].

The inconsistency of both TEX tests of strength and endurance should not be surprising as hip extensor deconditioning appears to not be associated with CLBP[91] and, as explained, without appropriate restraint of the pelvis the musculature of the hip extensors will serve to confound results. However, despite hip extensor deconditioning having apparently little association with LBP it seems some other aspect of TEX, perhaps ILEX, may be associated with it. As TEX is composed of both hip and lumbar extension it, therefore, seems logical that tests should attempt to remove the involvement of the hip extensors to examine ILEX. Thus, of key importance and inclusion to this review are studies that have used appropriate methods of testing ILEX. As shall also be noted, previous surgery may have implications for the results of studies examining the deconditioning hypothesis[92–95] and ideally participants with prior surgery should be excluded. However, only one study utilizing ILEX has controlled for this factor, yet may suffer from its own shortcoming of small sample size. Thus, in the following section all ILEX studies have been examined with this limitation noted.

Isolated lumbar extension studies

The validity of the extension test used is of great importance in examining the association between strength, endurance, and LBP, therefore studies that have considered this are potentially more useful in answering the question of whether specific lumbar extension deconditioning is associated with LBP. Unfortunately, in comparison with studies of TEX, studies of ILEX are relatively scarce. However, studies utilizing testing that appropriately restrains the pelvis consistently report significantly reduced ILEX strength in symptomatic CLBP participants compared with asymptomatic controls[96–98]. Other studies[99–102] have further reported reduced strength results from symptomatic participants compared to normal values obtained from healthy asymptomatic controls in other research[53].

There is, however, only one study by Lariviere *et al.*[40] of ILEX using valid restraints that does not support the link between specific deconditioning and CLBP. Lariviere *et al.*[40] reported no difference between asymptomatic and symptomatic CLBP participants in strength reported as maximum torque, or endurance reported as repetitions performed at a load equal to 60% maximum voluntary contraction (MVC). They commented, however, in discussion that the small sample size ($n = 18$) used may have meant a lack of the typical multifactorial heterogeneity in their non-specific CLBP group, potentially impacting the generalization to CLBP of this aspect of their research. It may also have resulted in a type II statistical error (i.e., failure to reject the null hypothesis). Other larger studies that have supported the link between ILEX weakness and CLBP have used in some instances upwards of 100 symptomatic participants and demonstrate reduced strength compared to healthy norms[99,100]. In addition and in particular, the study by Nelson *et al.*[99] of 895 CLBP participants suggested that a range of diagnoses existed in their sample (patients' diagnoses included non-specific CLBP, degenerative disc/arthritic disease, lumbar disc syndrome or spondylolisthesis/spondylolysis) and thus was likely quite representative of the typical heterogeneity of CLBP populations. Age, stature, and body mass were also similar between groups in the study by Lariviere *et al.*[40], however this was also reportedly the case for a number of other studies supporting the association[96–98], and so is unlikely to explain the difference in results.

One limitation of studies supporting the link between reduced ILEX strength and LBP is that these studies either did not report whether they excluded[97,99–102] or chose not to exclude[96,98] participants who had undergone previous surgery. Lariviere *et al.*[40] did exclude those having undergone previous lumbar surgery and, thus, this may explain the different results found by these investigators. As has been noted, previous surgery can have potentially deleterious consequences to the lumbar extensor musculature anatomy[92–95] and so might be thought to interfere with ILEX strength in symptomatic participants. Although a number of TEX studies have excluded those with previous surgery, with some supporting[37,74,77–82] and some refuting[88,89] an association between deconditioning and LBP, we must consider the inherent limitations of this approach already highlighted when specifically concerned with the lumbar extensors. There is certainly potential for further research to clarify whether differences in ILEX strength do indeed exist independent of previous lumbar surgery.

A final concern is the lack of statistical comparison with healthy control groups in some studies[99–102]. Although these results are consistent with those that have conducted statistical comparisons[96–98], this is a weakness and again something to be ensured in future research.

It is noted that though only one study reported upon tests of isolated lumbar extension endurance[40]ever, due to the inherent relationship between strength and endurance, it seems logical that the reported reduced lumbar extension strength in CLBP would be indicative of a reduced endurance also. The limitations discussed above also apply to this aspect of the study, however, and there is further scope for research specifically examining this.

Summary of strength and endurance studies of the lumbar extensor musculature

Of the studies examined, those employing sufficient pelvic restraints as their means of assessing lumbar extension have consistently reported results that lend support to the association of specific lumbar extensor deconditioning with CLBP[96–102], with only one exception[40]. It seems clear that when valid testing of ILEX is used, most evidence suggests a link between specific lumbar extension deconditioning and CLBP. However, it is unclear from purely this area of research whether this may in fact be due to the presence of previous surgery. Studies controlling for this factor utilizing a larger sample size should be conducted to further test this. Table 1 summarizes the findings of these studies.

If it is the case that deconditioning exists independent of previous surgery, a number of possible explanations may exist for the apparent association between ILEX deconditioning and CLBP: pain, anticipation of pain or pain avoidance behaviors, interfering with trunk muscle function; lack of motivation in asymptomatic participants; even deliberate malingering in some cases. Studies as early as those of McNeill *et al.*[62] suggested that the reduced TEX strengths seen in symptomatic compared with asymptomatic participants are most likely explained by the participants' avoidance of either large tensions in the posterior soft tissue or large compressive force on the lumbar motion segments. This conclusion would seem reasonable, being that there was an absence of studies of the lumbar extensor musculature in CLBP showing *in vivo* the condition of the lumbar musculature at the time of the study by McNeill *et al.*[62], to corroborate with the empirical findings on function. Indeed, strength is a product of both muscular force and the moment arm about which it acts, but the measurement of strength is significantly affected by volitional exertion. The concerns of McNeill *et al.*[62] were well justified in the absence of evidence specifically implicating muscular deconditioning *in vivo*. Evidence that has subsequently examined this, however, provides important information regarding the presence of specific lumbar deconditioning of the lumbar extensors in LBP. As such the next section shall detail and discuss this evidence.

Imaging and histochemical studies of the lumbar extensor musculature in LBP

Considerations for imaging and histochemical studies of the lumbar extensor musculature

As suggested, the data on reduced ILEX function should be further corroborated with studies specifically examining the lumbar extensor musculatures condition *in vivo*.

Table 1. Summary of studies testing strength and endurance of the lumbar extensor musculature in LBP.

Reference	Participants	Testing	Results	Comments
Lariviere et al.[40]	Healthy controls without LBP lasting 1 week in previous year, $n=18$ CLBP patients, $n=18$	Isolated lumbar extension MVC and number of repetition to failure at 60%MVC using customized dynamometer	No significant difference between groups	Those with previous lumbar surgery were excluded Age, body mass, height, BMI, body % and physical activity levels were similar between groups
Cassisi *et al.*[96]	Healthy controls without history of LBP, $n=12$ CLBP patients, $n=21$	Isolated lumbar extension MVC using MEDX	Lumbar extension significantly weaker in CLBP ($p=0.01$)	13 CLBP patients had undergone previous surgery although no effect upon lumbar extension strength was observed Age and height were similar between groups although body mass was greater in CLBP group
Holmes *et al.*[97]	Healthy geriatric female controls, $n=20$ CLBP geriatric female patients, $n=18$	Isolated lumbar extension MVC using MEDX	Lumbar extension significantly weaker in CLBP ($p<0.05$)	Age, height and weight were similar between groups
Robinson *et al.*[98]	Healthy controls, $n=12$ CLBP patients (53% having had previous surgery), $n=16$	Isolated lumbar extension MVC using MEDX was performed and 60% MVC determined at full extension for further EMG analysis during isotonic trial (see Table 3)	Absolute load used during isotonic trial was significantly lower in the CLBP group compared with the asymptomatic controls ($p<0.05$)	10 CLBP patients had undergone previous surgery Age, height and weight were similar between groups
Nelson *et al.*[99]	CLBP patients, $n=895$	Isolated lumbar extension MVC using MEDX	CLBP baseline data was compared graphically to healthy norms from Graves *et al.*[53] and shown to be considerably weaker	Patients diagnoses included non-specific CLBP, degenerative disc/arthritic disease, lumbar disc syndrome or spondylolisthesis/spondylolysis
Mooney *et al.*[100]	Strip mine workers (90% reported prior LBP), $n=197$	Isolated lumbar extension MVC using MEDX	Baseline data was compared graphically to healthy norms from Graves *et al.*[53] and shown to be considerably weaker	
Mooney *et al.*[101]	Healthy controls, $n=8$ CLBP patients, $n=8$	Isolated lumbar extension MVC using MEDX	CLBP baseline data was compared graphically to both healthy participants in the study and healthy norms from Graves *et al.*[53] and shown to be considerably weaker	Patients showed evidence of degenerative disc disease
Boyce *et al.*[102]	Small manufacturing plant workers (53% reported LBP), $n=20$	Isolated lumbar extension MVC using MEDX	Baseline data was compared graphically to healthy norms from Graves *et al.*[53] and shown to be considerably weaker	

Documentation of their roles in support and stability of the lumbar spine has motivated a large body of research examining their anatomical and histochemical condition in relation to LBP. Broadly, these studies can be divided into those that have examined the gross anatomy of the lumbar musculature (using imaging study, i.e., magnetic resonance imaging [MRI] or computed tomography [CT]) and those that have examined the histochemical nature (through use of muscle biopsy) or 'microanatomy' of the lumbar musculature.

Here we will review both, yet whilst doing so consider the many factors that may affect and limit the conclusions that can be drawn from these studies. Surgery via posterior approach can result in alteration of the lumbar musculature[92] which can be lasting[93,94]. However, there is evidence that only gross surgery, such as that for disc herniation, has this effect and that laminectomy and nucleotomy does not impart this damage to the musculature[95]. This is an important factor when considering the population examined. In some cases participants undergoing acute surgery have been examined and this presents an issue with determining whether deconditioning was present before surgery or is merely a result of surgery; indeed both may be the case[92].

Another issue that is involved in studies that have drawn bilateral (i.e., left and right) comparisons for evidence of asymmetry, or multiple vertebral level comparisons, is lack of asymptomatic controls. If deconditioning is present more on one side than the other, or more confined to a particular vertebral level, it is often considered that atrophy is local to symptoms[103]. However, without an asymptomatic group to compare it is impossible to say whether the asymptomatic side of symptomatic participants is normal or indeed atrophied itself, although to a lesser degree than the symptomatic side.

Additionally, age significantly impacts upon muscle degeneration[104–106]. Research that has compared symptomatic participants to age-matched asymptomatic controls is more valuable in determining the association of deconditioning with LBP. Other considerations include the validity of semi-quantitative analysis of images[107] and the value of measuring cross-sectional area (CSA) as compared to muscle density or fatty infiltration in imaging studies[74].

An issue with many of the studies examining so called 'normal' or 'asymptomatic' muscle histochemistry is that they have utilized biopsy samples from autopsy[108] or from acute disc herniation patients undergoing surgery[109,110]. This is justified by the assumption that short duration of spinal dysfunction would have little impact upon muscle condition[110], and early suggestions are that surgical procedure has little impact upon muscular condition due to asymmetric differences being unrelated to the side of herniation[109]. Due to the possible association between deconditioning and the initiation of LBP, these disc herniation surgery studies may perhaps be more indicative of the typical muscle condition that predisposes LBP development if it is known that the biopsies were taken before surgery began; however, this is often not reported. It is important to remember that gross surgical procedures such as those for disc herniation have themselves been shown to have an impact upon the musculature[95].

As a result of these concerns, our discussion focuses upon studies that have appropriately controlled for these factors (i.e., exclusion of previous surgery, control of age between groups). In addition, consideration of the potential for the presence of either deconditioning confined to a particular side or vertebral level will be compared to the potential for a general deconditioning. As with the TEX studies, those studies that have not considered such methodological factors as highlighted in this section have been summarized in the Supplementary Appendices provided and appear to show inconsistent associations both for imaging[75,76,94,101,103,106,111–119] and histochemical studies[92–94,106,109,120–126].

Imaging studies of the lumbar musculature

Firstly, the imaging studies that have examined the gross anatomy of the lumbar musculature will be reviewed. As noted there are numerous studies on this topic that have not controlled for the potentially confounding factors of age and previous lumbar surgery[75,76,94,101,103,106,111–119]. Of those examined for this review a handful of studies[74,91,107,127,128] did control for these factors and the results of them are summarized here (note that, although some studies have examined the psoas also, this review focuses upon the lumbar extensors).

Several studies have examined the CSA of either the paraspinal muscles as a whole group[91,127], the ES muscle group[74,127], and in one the quadratus lumborum (QL; which can initiate lumbar extension when bilaterally contracted)[91]. Kamaz *et al.*[91] examined absolute total paraspinal muscle CSA and found significant reduction in CLBP participants at the lower level of L4 but not at the upper. There was, however, a significant reduction in CSA of the QL at the upper level. Danneels *et al.*[127] found no difference in normalized ES CSA between symptomatic or asymptomatic CLBP participants. In addition, however, they also examined total paraspinal muscle CSA and did report significantly reduced CSA in CLBP participants. Comparing these results is problematic as Kamaz *et al.*[91] did not normalize their values. Although Danneels *et al.*[127] did not find a reduction in CSA of the ES, they did in the MF and attributed the reduction in total paraspinal CSA to the reduction in the CSA of MF.

Another study by Hultman *et al.*[74] comparing participants with intermittent LBP, CLBP, and healthy age-matched controls found no difference between groups for ES CSA. They did, however, find a significant reduction in ES density in the CLBP group. This study potentially brings the value of CSA as a sole measure of deconditioning or atrophy into question and may explain some of the disparity in results of other studies. Indeed, as in other physiological measures the absolute measurements of a particular variable (i.e., CSA, in a similar vein to mitochondrial volume[129]; or capillary density[130,131] that have been discussed elsewhere[132]) are often less valid than relative measures of density, and the same might apply to muscle CSA and muscle density. Muscle density may, therefore, be more representative of muscle atrophy as changes such as fatty infiltration may serve to maintain absolute CSA but would indicate that muscle density had indeed reduced. Yet, in light of this, Mengiardi *et al.*[107], when comparing age-matched controls, found no difference in ES fat percentage. Studies examining muscle density/fat content are expanded upon below.

CSA of the MF has also been examined by two studies, both of which support a link between reduced MF CSA and CLBP[91,127]. Both Kamaz *et al.*[91] and Danneels *et al.*[127] demonstrated that MF CSA was significantly reduced compared with healthy age-matched controls. Kamaz *et al.*[91] found these results consistent at both the upper and lower level of L4, and Danneels *et al.*[127] at just the lower L4 level.

When the ES has been examined for differences in muscle density or fat content between asymptomatic and symptomatic participants there have been contrasting findings. As noted, although they did not find any evidence of reduced CSA, Hultman *et al.*[74] noted significant reduction in muscle density of the ES in CLBP participants. Danneels *et al.*[127], however, also found no difference in ES muscle CSA without fat between their age-matched groups, and so they suggested that fatty infiltration may be more closely associated with age than indicative of atrophy[104–106]. Mengiardi *et al.*[107], when comparing age-matched controls, also found no difference in longissimus fat percentage using both a quantitative and semi-quantitative analysis.

In considering the MF, Danneels *et al.*[127] also reported no difference in muscle CSA without fat for the MF, despite an overall reduction in MF CSA. In contrast, however, Mengiardi *et al.*[107] did find a significant difference in fat percentage of the MF when considering the results of their quantitative analysis. The results of their semi-quantitative analysis, however, found no difference, and it seems reasonable to suggest that this lack of difference may stem from observer error.

Moderate differences between studies may perhaps be explained by the level of measurement. It has certainly been suggested and evidenced by some that atrophy of the ES or MF may be dependent upon level and side of symptoms due to denervation atrophy[103,115]. However, other evidence has suggested that a general atrophy at all levels and both sides may exist in CLBP participants compared with controls, and that asymmetry is merely more pronounced in those with specific symptoms of radiculopathy[117]. Although age was controlled in one study between groups[103] and participants with previous surgery were excluded from the other two[115,117] the lack of control of one or the other between these studies renders difficulty in concluding whether deconditioning and atrophy is level or side-specific or whether it is indeed more general. This is certainly an area requiring further research, the results of which may have important implications for prevention and treatment through use of exercise, particularly considering the different approaches used to address these, i.e., resistance exercise or motor control training.

Although not comparing asymptomatic and symptomatic participants, one other study is worthy of mentioning which did control for the influence of age and previous surgery. Kang *et al.*[128] examined CSA, both absolute and normalized to disc CSA, and used semi-quantitative analysis of fat content of the ES and MF in a group of CLBP participants as controls and a group of CLBP participants with degenerative kyphosis preparing to undergo corrective surgery. Although they were not able to compare their groups to healthy asymptomatic controls they did note that reduced CSA, both absolute and normalized, and more severe fat content, were found in the kyphosis group compared with the controls, and regression analysis showed MF atrophy to be most strongly associated. These results are interesting considering the influence of the musculature upon spinal stability and certainly present an area of future research to examine whether the general atrophy often seen in CLBP is more severe in light of structural dysfunctions, and whether this is causative or instead a result of these more severe conditions.

Although disparate, perhaps due to methodological issues, those studies reviewed (having excluded previous surgery and controlling for age) all suggest some form of deconditioning and atrophy, either reduced CSA, reduced density, or fatty infiltrations, being present in both the ES and the MF in CLBP participants compared with asymptomatic controls[74,91,107,127]. Considering that both play important roles in lumbar spine stability[23], this is potentially evidence for a plausible role of deconditioning in LBP.

Histochemical studies of the lumbar musculature

Imaging studies offer valuable insight into the gross anatomical condition of the musculature. Histochemical studies on the other hand are able to provide further detail considering individual fiber size, density, and differentiation between differing fiber types, as well as identifying specific pathological changes such as presence of small angulated fibers, target/core targetoid fibers, and fiber type grouping or group atrophy[7]. Mannion[7] has reviewed and highlighted a number of important findings from this area. She concludes by highlighting the difficulty of distinguishing cause and effect of fiber type characteristics (i.e., whether the observed characteristics existed prior to onset of LBP, or were a consequence of the presence of LBP). Her review also discussed fiber characteristics in relation to electromyographic (EMG) analysis of the lumbar spine musculature, and this will be discussed further in the following section. Here we will further consider the findings reported by Mannion[7], along with more recent findings. Again, studies have considered both the ES[77,94,106,109,120,121,133] and MF (both deep and superficial)[92,93,109,122–126].

When it has been made clear in studies that biopsies were taken before surgery, then the direction of the association between deconditioning and LBP might be better identified. However, biopsies are frequently taken during surgery or this is often not clarified[93,94,109,120–123,125,126]. Where it is not specified it is instead prudent to assume that the biopsy was taken during the operation, meaning we need to treat the results from these studies with caution. One study has shown that pathological changes are present before surgery, although further denervation is apparently caused by surgery, as shown in biopsies taken afterward[92], which certainly suggests that deconditioning may be present before surgery is initiated and, thus, associated with conditions for which surgery is recommended.

Studies of the histochemical condition of the paraspinal muscles in symptomatic CLBP participants that have not undergone surgery suggest the presence of fiber atrophy, pathological changes, and fiber type ratio alteration[106,124]. Neither of these cited studies, however, included asymptomatic controls. Zhao *et al.*[124] conducted bilateral comparisons and suggested that different findings between sides were affected by location of herniation, and that differences existed among those with central, bilateral, and unilateral pain. Prospective evidence has suggested that herniation can cause change in muscle activity, which might cause denervation atrophy[134]. Again, however, the absence of an asymptomatic control group renders the same difficulty as in other studies[103,115] when drawing conclusions (i.e., it is not known if the side without herniation was atrophied also).

Only one study has been conducted in the absence of the potential confounding influence of surgery, has controlled for the confounding effects of age, and also included a matched asymptomatic control group[77]. Crossman *et al.*[77] reported no difference in fiber size or fiber ratios between participant groups, and that both had a predominance of type I fibers. This is in contrast to Mannion's[7] earlier conclusions that symptomatic participants have a higher proportion of type IIX fibers. However, Crossman *et al.*[77] did not note the specific location of their biopsy sample and, thus, it is not clear whether these results refer to the ES, MF, or the paraspinal musculature as a whole.

Summary of imaging and histochemical studies of the lumbar extensor musculature

Although evidence suggests that deconditioning is indeed present in some form in symptomatic participants there is considerable disparity in methodologies in both imaging and histochemical studies. Data from imaging studies appear more consistent in their findings of some form of atrophy[74,91,107,127], as opposed to those from histochemical studies; however, only one histochemical study has controlled for previous surgery and age[77]. Indeed, although general deconditioning may be present in LBP, the findings of Crossman *et al.*[77] suggest that dominance of an adverse fiber type is perhaps not. Table 2 summarizes these studies.

Crossman *et al.*[77] also suggest that the differences in functional tests between asymptomatic and symptomatic participants' strength/endurance discussed in the previous section may, therefore, be due to the influences of psychological disturbance or motivation. However, we should consider that it is not only fiber type distribution that influences fatigue resistance but also capillary density, enzymatic activities, and associated metabolic processes[7,132]. So it is unsurprising that there is not a distinct relationship between this single variable and its associated end effect. Mannion[7] highlights that, because functional tests (i.e., strength/endurance) can be confounded by psychological disturbance, EMG should be employed to circumvent this and record more objective indices of muscle activation and fatigue. Indeed this measure might be considered to account for the many factors influencing fatigue due to its ability to accurately predict it[135] and that it also has a close association with physiological indicators of fatigue[136–138]. Cooper *et al.*[139] have shown that greater EMG amplitude increases occur during a test to fatigue in symptomatic participants (both surgical and non-surgical, suggesting a similarity underlying the two groups) compared with asymptomatic participants, and suggested that it indicated an increased central drive secondary to muscle wasting or denervation. Thus, EMG and other activation studies, therefore, may provide further insight into the deconditioning hypothesis and LBP.

Evidence suggests reduced strength/endurance in symptomatic participants which is further corroborated with *in vivo* evidence of muscular deconditioning being present. Further, and in consideration of the aforementioned concerns with participant effort in functional tests, the following section will complete the triumvirate of areas covered in examining deconditioning of the lumbar extensor musculature in LBP by reviewing studies that have employed EMG to assess fatigability.

Electromyography studies of fatigue in the lumbar extensor musculature in LBP

Considerations for electromyographic fatigue analysis of the lumbar extensor musculature

In consideration of the effect that deconditioning, and thus fatigability, may have on LBP, EMG has been used to attempt to control for influence of psychological disturbance or participant motivation on functional measures of endurance[7]. Thus, the information presented by these studies is also useful in examining the deconditioning hypothesis by corroborating evidence from the prior two sections which may support a link between lumbar extensor deconditioning and LBP. EMG can be used to interpret muscle activation and muscle force[140], but can also be used to more objectively demonstrate fatigability[7,135]. EMG is limited in many regards by such confounding factors as cross-talk (readings from synergist muscles), depth of active motor units from surface electrode, amplitude

Table 2. Summary of imaging and histochemical studies of the lumbar extensor musculature in LBP.

Reference	Participants	Testing	Results	Comments
Imaging studies				
Hultman *et al.*[74]	Healthy controls without history of LBP, $n=24$ Patients with intermittent LBP, $n=40$ CLBP patients, $n=21$	CSA and density of erector spinae using CT at L3 level	Muscle density was significantly lower in CLBP patients compared to both other groups ($p<0.05$) CSA did not significantly differ between groups	Those with previous lumbar surgery were excluded Age, height, body mass, and body composition similar between groups
Kamaz *et al.*[91]	Healthy controls without LBP or leg pain, $n=34$ CLBP patients, $n=36$	CSA of total paraspinal, multifidus, quadratus lumborum, psoas, and gluteus maximus muscles using CT at L4 upper and lower plates	CSA was significantly reduced in only paraspinal and multifidus at the lower plate in CLBP ($p<0.01$) CSA was significantly reduced in only multifidus, psoas, and quadratus lumborum at the upper plate in CLBP ($p=0.05$) No significant differences between CSA of gluteus maximus	Those with previous lumbar surgery were excluded Age and BMI similar in both groups.
Mengiardi *et al.*[107]	Healthy controls without history of LBP in previous 2 years, $n=25$ CLBP patients, $n=25$	CSA of multifidus and longissimus fat content and semi-quantitative grading using 5 classification system using MRI at L4-L5 level	CLBP patients showed significantly greater fat content in the multifidus ($p<0.05$) No difference found using semi-quantitative system	Those with previous lumbar surgery were excluded Age, sex, and BMI matched between participant groups
Danneels *et al.*[127]	Healthy controls without history of previous LBP, $n=23$ CLBP patients, $n=32$	Total CSA and muscle CSA of total paraspinal, erector spinae, multifidus, and psoas muscles using CT at upper L3, and upper and lower L4 normalized	Total CSA of paraspinal and multifidus muscles significantly smaller at lower L4 in CLBP ($p<0.05$) No significant difference for erector spinae or psoas	Those with previous lumbar surgery were excluded in addition to those who had participated in training for the lower back muscles in the previous 3 months Age, height, weight, and activity similar between groups
Kang *et al.*[128]	CLBP patients with lumbar degenerative kyphosis undergoing corrective surgery, $n=54$ CLBP control patients, $n=54$	CSA and muscle to disc CSA ratio of psoas, erector spinae, and multifidus was assessed at L4/L5 level and fatty infiltration of psoas, erector spinae, and multifidus assessed at L3/L4 using three grade classification using MRI	CSA and muscle to disc CSA ratios for all muscles were significantly lower in the lumbar degenerative kyphosis group compared with controls ($p<0.001$) with regression analysis showing multifidus wasting to be most strongly associated ($p<0.001$) Severe fatty infiltration was significantly more common in lumbar degenerative kyphosis compared to CLBP controls ($p<0.05$)	No healthy control group for comparisons Those with previous lumbar surgery were excluded from CLBP control group Age- and sex-matched between groups and symptom durations were similar Body mass and BMI was significantly higher in CLBP controls No difference in degenerative changes (degenerative disc disease, herniation's, stenosis, or spondylolithesis) between groups
Histochemical studies				
Crossman *et al.*[77]	Healthy controls without LBP lasting >3 days in previous 12 months, $n=32$ CLBP patients, $n=35$	Percutaneous biopsy of paraspinal muscle (specific location not noted) for fiber CSAs and fiber typing	No significant differences between groups for any fiber histochemical comparisons	Those with previous lumbar surgery were excluded Age, gender, and all anthropometric characteristics similar between groups

related to motor units and muscle fiber-types, variable firing rates, muscle-fiber length, velocity, and contraction type[141–146]. Cross talk is of particular issue when differentiating specific lumbar extensor musculature[147]. However, in considering power spectrum analysis of rate of change in EMG spectral variables[148] (i.e., root mean square amplitude, mean, median, or mode frequency slopes, initial frequencies, etc.) for determining fatigability these might perhaps not be so confounding an issue as they would presumably remain constant systematic errors while such EMG parameters would change with fatigue.

When looking at LBP populations we should consider that the MVC-normalized EMG signal amplitude measured may perhaps be influenced by insincere effort[149,150]. Roy *et al.*[151], however, have shown that EMG measures of fatigability provide accurate classification independently of MVC, suggesting their greater objective power in discriminating between symptomatic and asymptomatic groups compared to simply measuring relative activation levels.

Although it seems EMG measures of fatigability are more valid, a point must be considered when interpreting their results; that is, whether participants performed exercise to momentary muscular failure (MMF; i.e., maximal intensity of effort)[32,33,152]. These studies should not surprisingly show a difference in fatigue indices from start to end of exercise performance within all groups participating in testing, but presumably would show no difference in between-group comparisons as both will be maximally fatigued. Change in fatigue indices over a fixed number of repetitions or time (i.e., as an *absolute* measure) would instead be the most appropriate means of detecting fatigue-related differences between symptomatic and asymptomatic groups, and, considering the issue with-normalizing to MVC in LBP participants, should preferably utilize an absolute load. In some studies the absolute load utilized has been the participant's torso mass during TEX. Thus, an important consideration for between-group comparisons of fatigue during TEX is whether body or torso mass was similar.

Geisser *et al.*[153] have conducted a meta-analysis of the use of trunk surface EMG comparing asymptomatic and symptomatic participants, and comment that EMG recordings from non-maximal tasks are likely to be more reliable than those involving maximal exertions. However, we should remember that both absolute and relative amplitude levels will be subject to the aforementioned limitations of EMG including an insincere effort, whereas fatigue may not be. EMG measures of fatigability should objectively quantify fatigue independently of MVC[151], where a significant change in fatigue is unlikely to be seen if insincere effort is put forth. Geisser *et al.*[153] reported an effect size of zero for EMG measures of fatigability during isometric trunk exertions, suggesting no difference between symptomatic and asymptomatic participants. However, a difficulty lies in interpreting these results, partly due to the methodological differences of studies included, but also because Geisser *et al.*[153] do not clarify EMG locations and whether extensor or flexor musculature, or a combination of the both, was being examined. Nor do they comment in more detail on the intensity of effort of the activity (i.e., whether it was performed to MMF or to an absolute time/number of repetitions). Their meta-analysis included seven studies[37,78,87,98,135,154,155] examining EMG measures of fatigability in LBP; however, a number of studies also examining fatigue indices that were present at the time of its publication were not included[151,156–159]. Being that EMG measures of fatigability are more valid and applicable to our present discussion of the deconditioning hypothesis we will further examine the studies analysed by Geisser *et al.*[153] along with those not included in their analysis, as well as further studies that have been conducted more recently[40,79–81,88].

For sake of clarity in this review, although numerous methods of analyzing the EMG signal for determination of fatigability exist between studies, here these methods are collectively referred to as EMG 'fatigue indices', as a critical comparison of the specific methods of analysis is beyond the scope of this review[160]. Due to the difficulty of cross talk between the paraspinal musculature when using surface EMG[147], we do not attempt to differentiate between, for example, the ES or MF, and instead consider the studies reviewed to offer information regarding the lumbar extensor musculature as a whole. Being that, as previously noted, surgery can have considerable confounding effects upon the lumbar extensor musculature[92–95], we have focused in this section upon those studies which have controlled for this[77,88,135,154]. Again, as with previous sections those studies excluded from discussion (in this case those not controlling for surgery or those which have had participants perform exercise to MMF[37,40,78–81,87,98,151,155–158]) have been summarized within the Supplementary Appendices.

Fatigability studies of the lumbar musculature

The studies reviewed utilizing measurements of EMG fatigue indices have examined differences between asymptomatic and symptomatic participants using different methods. Some have used both discriminant analysis and regression to identify whether such measures can successfully classify participants, and others have drawn simpler between-group comparisons.

Roy *et al.*[135] have performed several studies examining fatigue indices in LBP, one of which controlled for both factors noted. They examined fatigue indices during 60-s standing isometric TEX contractions at 40%, 60%, and 80% MVC. Discriminant analysis correctly classified between asymptomatic controls and symptomatic CLBP participants at 40% MVC (92% controls, 82% CLBP) and 80% MVC (84% controls, 91% CLBP); however, results were less favorable at 60% MVC (67% controls, 75% CLBP; a later study by Peach and McGill[155] clarifies this anomaly, although it should be noted they do not note whether those with previous surgery were excluded). This study also looked at two-level analysis (Lumbar level and %MVC level), finding that fatigue indices at L5 for 80% MVC showed the most favorable classification (75% controls, 75% CLBP).

In an early study by Mayer *et al.*[154], participants performed an isometric TEX hold using a roman chair in the same manner as the Biering-Sorensen test. Participants performed a series of 10 holds for 15-s each with a rest period of 10-s between holds. Between-group comparisons of fatigue indices for both the first five holds, as well as the full 10, demonstrated significantly greater fatigue in the symptomatic CLBP group than in the asymptomatic controls before completion of an intensive rehabilitation program. After the program the difference between groups was reduced and still significant for the 10 holds, yet there was no significant difference when data for five trials were compared.

Humphrey *et al.*[79] considered a range of fatigue indices calculated from power spectrum analyses during a back lift test. They reported significant differences in fatigue indices between CLBP participants and controls. In addition they reported that logistic regression showed high sensitivity and specificity in classifying CLBP participants. However, although the variables considered could discriminate between symptomatic CLBP participants and asymptomatic controls there were varying degrees of accuracy. They noted that those variables that could potentially be affected by load (peak amplitude, median frequency) were less accurate as predictors; however, load independent variables (such as initial median frequency and half width) offered a higher degree of accuracy. Humphrey *et al.*[79] also included a group of participants with a past history of LBP. No variables were able to discriminate these from either the CLBP participants or the controls, although this was suggested to be due to the comparatively small sample size for this group (healthy controls $n = 175$; CLBP participants $n = 145$; past history participants $n = 30$).

A later study by Da Silva *et al.*[88], however, offers contrasting results. They found no significant difference between asymptomatic controls or symptomatic CLBP participants in fatigue indices between groups for 60-s contractions at 50% MVC for the Biering-Sorenson test, a standing extension test, and also a semi-crouching back lift test. It is unclear as to the reason for this contrasting finding; however, Da Silva e*t al.*[88] suggest that the CLBP group studied may not have been sufficiently impaired to demonstrate a difference based upon the low results from the Oswestry disability questionnaire (~12%).

Another study by Crossman *et al.*[77] has also reported no difference in fatigue indices between healthy asymptomatic and symptomatic CLBP participants during standing TEX using 60% MVC for 60-s and also during the Biering-Sorenson test. Crossman *et al.*[77] comment that this may perhaps not be surprising due to the lack of differences in histochemical analysis of fiber typing in their participants. However, they also note concerns regarding the loads used by CLBP during the 60% MVC TEX test in particular, speculating that these participants may not have given a sincere MVC and, thus, they may have been using <60% MVC during this test. As noted earlier MVCs have been evidenced to be affected by this[149,150] and as a result we noted that the use of an absolute load may be of greater validity in determining differences in fatigability. Crossman *et al.*[77] did also have participants perform the Biering-Sorenson TEX test, also reporting similar body mass between groups, which suggests that the absolute loading between groups for this test was similar. However, this test was performed to MMF and so it is again unsurprising that no differences in fatigue were found. Yet, CLBP participants did demonstrate significantly lower endurance times and so, assuming they did perform the test to MMF (and that also healthy controls did), the lower endurance time might indicate greater fatigability. However, it again must be noted that this is a test of TEX and so the endurance time is not specifically indicative of the lumbar extensors.

Unfortunately, considering the potentially confounding effect of relative load being influenced by insincere MVCs in CLBP participants, we are left with only the results of Mayer *et al.*[154] which do suggest greater fatigability in CLBP participants. It is difficult to discern whether any other factors may have affected the results between studies apparently supporting the presence of deconditioning through EMG fatigue indices[79,135,154] and those suggesting it is not present[77,88]. All have used a range of electrode placement sites (T10/L1/L2/L3/L4/L5), many in different combinations, a range of tests (standing TEX, prone TEX, back lift test), both relative and absolute loads as noted, and also a range of test timings (30-s, 60-s, and 10 repetitions of 15-s); as such it is unclear as to what effect these variables may have upon the study's findings.

Summary of electromyography studies of fatigue of the lumbar extensor musculature

In summary, of the studies reviewed, it appears that objective measures of fatigability show contrasting results. Those controlling for previous surgery and using standardized timed protocols show some evidence in support[79,135,154] and some against[77,88] the presence of deconditioning. One study that has also controlled for the potentially

Table 3. Summary of studies testing fatigability with EMG of the lumbar extensor musculature in LBP.

Reference	Participants	Testing	Results	Comments
Crossman *et al.*[77]	Healthy controls without lasting >3 days in previous 12 months, $n = 32$ CLBP patients, $n = 35$	EMG recorded bilaterally from lumbar paraspinal muscles at L4–L5 level during standing isometric trunk extension for 60 s at 60% MVC and during the Biering-Sorensen test	EMG fatigue indices were similar between groups for the Biering-Sorenson test and also the 60%MVC test	Those with previous lumbar surgery were excluded Age, gender, and all anthropometric characteristics similar between groups
Da Silva *et al.*[88]	Healthy controls without history of LBP in previous year, $n = 15$ CLBP patients, $n = 13$	EMG recorded bilaterally from lumbar paraspinal muscles at T10, L1, L3, and L5 levels during standing trunk extension and back lift at 50%MVC for 60 s, and during Biering-Sorenson test for 60 s	No difference in EMG fatigue indices between groups	Those with previous lumbar surgery were excluded Age, height, and weight were similar between groups
Roy *et al.*[135]	Healthy controls, $n = 12$ CLBP patients, $n = 12$	EMG recorded bilaterally from lumbar paraspinal muscles at L1, L2, and L5 levels during standing isometric trunk extension for 60 s at 40% MVC, 60% MVC, and 80% MVC	Discriminant analysis of EMG fatigue indices successfully classified 92% controls, 82% CLBP at 40% MVC, 67% controls, 75% CLBP at 60% MVC and 84% controls, 91% CLBP at 80% MVC	Those with previous lumbar surgery were excluded Age, height, and weight were similar between groups
Mayer *et al.*[154]	Healthy controls, $n = 11$ CLBP patients, $n = 10$	EMG recorded bilaterally from lumbar paraspinal muscles at L3 level 3 cm from mid-line during 10 isometric trunk extension holds on a roman chair lasting 15 s each and with 10 s rest between each hold	EMG indices of fatigue showed significantly greater fatigue in the CLBP group compared to controls ($p < 0.01$)	Those with previous lumbar surgery at level of EMG placement were excluded Age and torso weight similar between groups

influencing factor of sincere effort by CLBP participants does, however, suggest the presence of some degree of deconditioning[154]. Table 3 summarizes these studies.

Thus far it has been evidenced that there may be an association between measures of, and variables associated with, lumbar extensor deconditioning (i.e., reduced strength/endurance testing of lumbar musculature, deconditioning shown by imaging and histochemical examination of the lumbar musculature, and increased fatigability of the lumbar musculature shown by EMG fatigue indices) and LBP, hence providing support towards the deconditioning hypothesis. Theoretically, muscular deconditioning could lead to instability and altered joint biomechanics and thus result in injuries (either single macro-trauma or cumulative micro-trauma) which may instigate pain causing mechanisms. Therefore, we might expect that the presence of such deconditioning, whilst being a consistent association with CLBP, might also predict the development of LBP or incidence of low back injury in initially asymptomatic individuals also. Prospective studies have examined whether this is in fact the case, and the following section will discuss the evidence, implicating deconditioning's effect upon injury and development of LBP.

Prospective studies of lumbar extensor deconditioning in LBP

A concern with cross-sectional studies is that causation cannot be logically determined from association (it should also be noted that a lack of association does not necessarily imply a cause and effect relationship does not exist). Despite a consistent association being one of the criteria for causation as determined by Hill[161], and the consistency of some degree of deconditioning with CLBP, in addition to biological plausibility of which there is evidence implicating deconditioning, it still cannot solely be taken as evidence for a causative relationship, nor the direction of that causation. Prospective studies provide clearer indication for a temporal relationship between variables and allow us to consider whether the potential plausibility for deconditioning to actually lead to LBP can be evidenced.

Although in previous sections we have been selective over those studies discussed based upon methodological considerations highlighted, this section considers a more liberal range of literature. The reason for this is due to the relative paucity of prospective studies that have appropriately controlled for the factors previously highlighted in this review. Thus, the studies reviewed in this section should be considered tentatively and it is noted that further research is required to definitively test the deconditioning hypothesis and the presence of a prospective relationship between deconditioning and LBP.

Prospective evidence from strength and endurance in LBP

Biering-Sorensen[68] found that weak TEX was a predictive residual sign of recurrent LBP or CLBP over a 1-year follow-up, although did not significantly predict first time occurrence. A study by Leino *et al.*[69] indicated that there was little prognostic value of tests of dynamic TEX in predicting LBP over a 10-year period but suggested instead an effect of the latter upon the former (i.e., symptoms, or degree of symptoms at baseline, had prognostic value in predicting reduced trunk muscle function at follow-up). Disappointingly, however, Leino *et al.*[69] omitted dynamic TEX tests from their follow-up testing. Initial testing consisted of prone dynamic TEX efforts, while follow-up data are reported for standing isometric TEX efforts. This presents difficulty in interpreting the effect of LBP presence at base-line affecting TEX muscle function at follow-up as it is a case of comparing different tests to identify change[162]. This makes the conclusions questionable. The dynamic tests consisted of the number of repetitions performed over 30-s which might be considered more specifically a test of the ability to complete TEX movement quickly, not TEX strength/endurance *per se*. The isometric test on the other hand was indeed a TEX MVC and, thus, a measure of TEX strength. The data from Leino *et al.*[69] compare two entirely different tests with no clear conclusions being evident. It would seem that there was little difference in relative risk of LBP development when low and high performers in dynamic extension were followed up. Contrastingly Rissanen *et al.*[163], utilizing the same dynamic test, reported significant prediction of back disorder disability over an average 12-year follow-up. However, such dynamic TEX testing does not really offer an appropriate presentation of muscle function[164–166] and so interpreting the predictive results from this in light of the deconditioning hypothesis is questionable. It should also be pointed out that again tests of TEX are not specifically indicative of lumbar extensor muscle function.

A study by Newton *et al.*[167] using a more consistent study design of dynamic isokinetic TEX testing (i.e., utilizing the same test both at baseline and follow-up) in prospective evaluation of LBP development, however, also suggested that it held no predictive value. Despite this, prospective implementation of the same battery of isokinetic tests as a pre-employment fitness evaluation in order to place workers in appropriate job areas has been demonstrated to significantly reduce injury rates[168]. This suggests a potential connection between physical function and task demands in predicting injury and perhaps explains the lack of predictive value in these tests when this is not considered[167].

Batti'e *et al.*[169] reported that greater strength was actually a risk factor for report of back problems over a 4-year period. However, closer inspection of their results shows that this was only significant for arm and leg lift strength and that torso lift (TEX) was not significant. When their data were adjusted for age there were no significant correlations. Another prospective study has reported that a reduced trunk extension/flexion strength ratio is a significant risk factor for development of LBP over a 5-year period[170]. That extensor deconditioning may be more significant than flexor deconditioning in those with LBP has been highlighted in previous research[50,62,63,75,91,101,112,127] and it, thus, is interesting that a greater relative deconditioning of the extensors is shown to be predictive of future LBP. Kujala *et al.*[171], on the other hand, suggested that neither isometric nor dynamic TEX performance were predictive of first time LBP in addition to strength ratio being unrelated in their sample group. However, their results did indicate a significant effect of musculoskeletal loading as well as reporting that taller participants (who may experience greater loading due to a greater external TEX moment) were more likely to develop LBP. Thus, their results are somewhat supportive of the concepts conveyed by Reimer *et al.*[168], and also Chaffin *et al.*[172] and Keyserling *et al.*[173] using the same test battery as Batti'e *et al.*[169], in that strength relative to physical demands is important. So, although the population studied did not differ in their initial strength, those who engaged in heavier loading were weaker relative to their loading demands[171].

Associations between weak TEX and LBP have also been reported in younger populations[73,86,174]. The studies of Salminen *et al.*[73,174] involved a 3-year follow-up and showed weak TEX associated with LBP at baseline and follow-up. Despite this, there was no predictive validity of TEX in development of LBP at follow-up. Studies have, however, also examined adolescents, showing prospective associations between TEX weakness and development of LBP[170,175].

TEX endurance has also been used in prospective studies. Poor TEX endurance has been identified as a risk factor for LBP incidence in some studies[68,83,175]. However, one study's findings indicate that it has no use in predicting future LBP[84]. Gibbons *et al.*[84] note, however, that the difference in results between theirs and previous studies may be due to type II error. Their sample size ($n = 43$) for follow-up in incidence of LBP after initial testing was much lower than the sample used by Biering-Sorensen[68] ($n = 982$), and also the samples used by Luoto *et al.*[83] ($n = 126$) and Sjolie & Ljunggren[175] ($n = 86$), which might suggest that their data would be more likely to present a type II error (i.e., a failure to reject the null hypothesis) from a lack of statistical power through low sample size. Thus, the larger and more numerous studies do indicate the predictive potential of low TEX endurance in development of LBP.

Adams *et al.*[176] conducted a large prospective study examining physical factors; including TEX endurance and back lift test MVC and examined EMG fatigue indices over 20-s. Their results suggest back lift strength was not predictive of LBP, but TEX endurance time was. An earlier study by Mostardi *et al.*[177] that demonstrated no predictive value of strength also performed a back lift test. In spite of this, another larger study ($n = 1652$) has also employed the same back lift method amongst other fitness measures and found that there was significant predictive value between those with the lowest, middle, and best strength and fitness; the least fit sustaining the greatest proportion of low back injuries and the most fit sustaining the least[178].

Where many previous studies have used less valid measurements of lumbar function (i.e., TEX testing), another prospective study by Mooney *et al.*[179] examined low back injury rates and their relationship to ILEX strength. One hundred and fifty-two shipyard workers were tested for ILEX strength and followed up for 2 years. In this period 9% ($n = 14$) reported low back injuries; however only two of these had below normal ILEX strength. These injury rates (9%), however, are considerably less than those reported for many other US industries[180]. The majority of the workers tested in the study by Mooney *et al.*[179] had normal ILEX strength. Thus, the relatively low rates of injury for the participant sample as a whole actually suggest that normal strength may be protective and that the injuries that were sustained were potentially outliers. Indeed, of the injuries reported the highest incidence was within the heavy work categories and, thus, these injuries may have represented accidents during heavy work[9], whereby task demands exceeded physical function[168,171–173]; however, no further detail was reported on the nature of the sustained injuries.

Prospective evidence from MRI and EMG in LBP

Although most prospective studies have examined the role of deconditioning from a perspective of functional tests of strength and endurance, there have been others that have examined imaging tests of the lumbar extensor musculature as well as EMG fatigue indices of the lumbar extensors. Gibbons *et al.*[84], using MRI, examined CSAs, proton density weighted signals, and T2-weighted signal intensities of the ES, QL, and psoas major prospectively, yet found no significant predictive value from any of these variables. Participants who suffered from LBP during the follow-up period did show slightly higher signal intensities which might be indicative of greater fatty infiltration and thus muscular deconditioning. The fact that these were not found to be significant may be a result of a type II error again due to the sample size used ($n = 128$). Although similar to the higher sample sizes in studies of TEX endurance, the authors are not aware of any other prospective imaging studies and so, unlike the endurance tests, this cannot be compared and confirmed.

Adams *et al.*[176] also utilized EMG fatigue indices in their study, yet found that there was no predictive value over 3 years follow-up. Another study by Mannion *et al.*[159], however, reported prospective data for 200 young nurses who had never before suffered from serious LBP. EMG fatigue indices were recorded at baseline and followed up for 12 months. The result showed that greater fatigability significantly predicted development of first time LBP. Stevenson *et al.*[181] reported on a variety of variables included in a predictive model of LBP over a 2-year period, including EMG fatigue indices in the final model which were significantly predictive of LBP occurrence in the previous 6 months. Finally a study by Heydari *et al.*[182] has also examined EMG fatigue indices prospectively in 105 participants with no previous history of LBP. They also reported that greater fatigability was predictive of subjects self-rating of LBP at 2-year follow-up.

Summary of prospective studies

It seems that a number of prospective studies are suggestive of deconditioning as potentially etiological within development of LBP[68,84,170,174,175]. These studies have predominantly employed methods examining TEX strength, endurance, and trunk extension/flexion ratios and so, as highlighted in the discussion of studies examining strength and endurance, it must be considered that there are limitations to these methods. However, one study has prospectively examined ILEX, yet, due to the limitations of this study, and depending on perspective, its data could be interpreted either in support of or against lumbar extensor deconditioning as being causative in LBP development[179]. Evidence from other methods of examining deconditioning is contrasting. MRI shows no predictive value[84]; however, as discussed, there is a lack of other imaging studies to compare this result to. In contrast, it appears that EMG fatigue indices may be predictive of LBP development[159,181,182]. Thus, although disparate, there is certainly some prospective evidence to support the deconditioning hypothesis. Table 4 summarizes these studies.

Discussion

It would appear that there certainly exists evidence indicative of some role specifically of lumbar extensor deconditioning in LBP, which may be causative, yet there is certainly scope for improving earlier studies with more appropriate examination of this relationship.

Table 4. Summary of prospective studies of lumbar extensor musculature deconditioning in LBP.

Reference	Participants	Testing	Results	Comments
Biering-Sorenson[68]	Men aged between 30, 40, 50, and 60 years old, $n = 449$ Women aged between 30, 40, 50, and 60 years old	Biering-Sorensen test conducted at baseline 1 year follow-up with questionnaire concerning first time occurrence, recurrence, or persistence of LBP	First time occurrence was significantly associated with low endurance time	
Leino *et al.*[69]	*Baseline participants* Participants with 'Good' low back status, $n = 578$ Participants with 'Intermediate' low back status, $n = 260$ Participants with 'Bad' low back status, $n = 64$ *Follow-up participants* Participants with 'Good' low back status, $n = 239$ Participants with 'Intermediate' low back status, $n = 203$ Participants with 'Bad' low back status, $n = 210$	Standing dynamic trunk extension/flexion maximum repetitions performed over 30 s with buttock and thighs against a supporting plate and ankles tied by a belt conducted at baseline Standing isometric trunk extension/flexion MVC with buttock and thighs against a supporting plate and ankles tied by a belt conducted at 10 year follow-up in addition to questionnaire and assessment of low back symptoms and status	Trunk strength was not predictive of low back symptoms or status at follow-up	
Luoto *et al.*[83]	Healthy participants without history of LBP in previous year at baseline, $n = 167$	Biering-Sorensen test and questionnaire regarding previous and present LBP conducted at baseline 75% of participants were available for follow-up at 1 year with the same questionnaire, $n = 126$	Endurance time was significantly associated with first time occurrence of LBP when adjusted for age, sex, and occupation ($p < 0.05$) Endurance time broken into tertiles (poor, medium, good) showed a non-linear dose–response relationship with first time occurrence of LBP ($p < 0.04$) Relative odds ratio compared to 'good' for 'medium' and 'poor' were 1.4 (95% CI = 0.4–4.2) and 3.4 (95% CI = 1.2–10.0), respectively	
Gibbons *et al.*[84]	Healthy participants without history of LBP in previous year at baseline, $n = 43$	Isokinetic back lift MVC, psychophysical back lift test, Biering-Sorensen test, CSA, proton-density weighted signal, and T2-weighted signal of erector spinae, quadratus lumborum, psoas major, and total paraspinal muscle using MRI, and interview regarding previous and present LBP conducted at baseline Interviews regarding LBP were conducted at 1 year follow-up	Neither back lift, psychophysical back lift or endurance time differed between those with and without LBP at follow-up, nor where they associated with frequency of LBP at follow-up Neither CSA, proton-density weighted signal, or T2-weighted signal differed between those with and without LBP at follow-up, however, total paraspinal CSA, and proton-density weighted signal and T2-weighted signal of erector spinae, quadratus lumborum, psoas major were significantly associated with frequency of LBP at follow-up ($p < 0.05$)	

(continued)

Table 4. Continued.

Reference	Participants	Testing	Results	Comments
Mannion *et al.*[159]	Healthy nurses without history of LBP, $n = 200$	EMG recorded bilaterally from lumbar paraspinal muscles at T10 and L3 level 3–4 cm from mid-line during Biering-Sorenson test and maintenance of 80% MVC for 28 s at baseline Postal questionnaire regarding LBP conducted at 1 year follow-up	13% developed serious first time LBP during the follow-up period EMG indices of fatigue during Biering-Sorenson showed greater fatigue was significantly associated with development of first time LBP at follow-up ($p < 0.05$), however endurance time was not associated with first time LBP	
Rissanen *et al.*[163]	Participants from the Mini-Finland Health Survey, $n = 535$	Dynamic trunk extension/flexion maximum repetitions performed over 30 s with buttock and thighs against a supporting plate and ankles tied by a belt conducted at baseline Average 12 year follow-up to time until retirement due to work disablement, death, or end of observation period for primary diagnosis as cause of work disability	At follow-up of 56 incident cases 15 were due to back disorders Adjusted relative risks in multiple models showed trunk extension performance significantly predicted back disorder disability risk ($p = 0.04$–0.002)	
Newton *et al.*[167]	Healthy participants without history of LBP, $n = 70$	Isokinetic trunk extension, flexion, rotation, and back lift MVC and psychophysical lift conducted at baseline 1 year follow-up with questionnaire concerning first time occurrence, recurrence, or persistence of LBP	23% developed LBP during the follow-up period, yet at least 6 months after initial assessment in all cases None of the isokinetic measures differed between those who did and those who did not develop LBP	Those with previous lumbar surgery were excluded
Reimer *et al.*[168]	Healthy prospective order selector employees for 1989, $n = 122$ Healthy prospective order selector employees for 1990, $n = 122$ Healthy prospective order selector employees for 1991, $n = 122$	Dynamic lift capacity, isokinetic trunk extension, flexion, rotation, and back lift MVC and psychophysical lift conducted at baseline to determine placement in employment as an order selector in a warehouse grocery distributor 2 year follow-up with questionnaire concerning first time occurrence, recurrence, or persistence of LBP	After implementation of prospective evaluation for employment placement in 1989, incidence of low back injuries were significantly reduced by 32% in 1990 and 41% in 1991 ($p < 0.001$)	
Batti'e *et al.*[169]	Employees working for a large aircraft manufacturer ($n = 497$ reporting LBP in previous 10 years), $n = 2178$	Isometric MVC for torso, arm, and leg lift was conducted at baseline 4 year follow-up conducted for claims related to low back injuries or LBP	Participants with higher MVC for arm, leg, and torso lift were at higher risk for LBP and low back injury ($p = 0.01$, 0.03, and 0.26, respectively). When adjusted for age and sex, however no association was present	Due to an injury rate of 0.6% during torso lift testing it was discontinued. $n = 495$ participants completed torso lift testing, $n = 2158$ completed arm lift testing, and $n = 2102$ completed leg lift testing
Lee *et al.*[170]	Healthy student participants without history of LBP, $n = 67$	Isokinetic trunk extension, flexion, and rotation MVC conducted at baseline 5 year follow-up concerning LBP incidence	27% developed first time LBP during the follow-up period Ratio of extension/flexion strength at baseline was significantly lower in participants who developed first time LBP, ($p < 0.05$)	Age, height, weight, and smoking habits were similar between groups

Kujala *et al.*[171]	Healthy participants without history of LBP, $n=262$	Standing isometric trunk extension/flexion MVC was conducted at baseline 5 year follow-up with questionnaire was conducted regarding type, frequency, severity, and functional limitations of LBP	47% developed first time LBP during the follow-up period, 11% of these reporting it as being of monthly frequency, 17% reporting radiating limb pain, and 2% having been hospitalized due to LBP Trunk extension/flexion was not associated with development of first time LBP	Age, weight, and BMI were similar between groups Height, occupational physical demands, and occupational musculoskeletal loading were significantly associated with first time LBP ($p<0.05$)
Chaffin *et al.*[172]	Pre-employed plant workers in a variety of jobs involving manual lifting, $n=551$	Isometric MVC for torso, arm, and leg lift in addition to job-specific demands was conducted at baseline Preventative effectiveness of strength relative to job demands were evaluated by examining incidence and severity of low back injuries over an 18 month follow-up period Participants were grouped into tertiles relating to their individual strength relative to their job demands	As job strength requirements exceeded participant strength the incidence and severity of low back injuries increased at a ratio of 3:1 across the tertiles	
Keyserling *et al.*[173]	Pre-employed plant workers applying for a range of 20 varied jobs, $n=71$	Isometric MVC for torso and arm lift, and push in/out in addition to job-specific demands was conducted at baseline Preventative effectiveness of strength relative to job demands evaluated by placing of experimental ($n=20$) group into jobs matching strength, whereas control group ($n=51$) were not Incidence of musculoskeletal injuries were evaluated over a 1 year follow-up period	During the follow-up period the control group experienced 19 incidences of musculoskeletal injuries compared to 0 in the experimental group	Age, weight, and height were similar between groups
Salminen *et al.*[174]	Healthy children, $n=38$ Children with LBP, $n=31$ Children with LBP and sciatica, $n=7$	Biering-Sorensen test, sit up isometric test with knees at 90°, and MRI conducted at baseline 3 year follow-up period evaluating LBP ever, LBP in past 12 months, and recurrent/continuous LBP	Both flexion and extension endurance times were significantly lower in LBP groups ($p<0.05$) at baseline and follow-up, yet endurance time was not predictive of development of first time LBP	Age-, sex-, school-matched between groups
Sjolie & Ljunggren[175]	Healthy adolescents, $n=86$	Biering-Sorensen test and questionnaire regarding LBP conducted at baseline 3 year follow-up period with questionnaire was completed	High mobility/endurance time ratios were significantly associated with development of LBP at follow-up when adjusted for gender, LBP at baseline, and well-being and physical activity at follow-up (OR $=1.5$–1.9, 95% CI $=1.1$–3.2, $p<0.05$)	
Adams *et al.*[176]	Healthy nurses without history of LBP, $n=262$ Healthy nurses who had previously suffered with 'non-serious' LBP, $n=141$	Biering-Sorenson tests, isometric back lift MVC and back lift at 80% MVC for 20 s while EMG recorded from T10 and L3 conducted at baseline 3 year follow-up (every 6 months) conducted using questionnaire regarding LBP in previous 6 months	Endurance time at 3 year follow-up was significantly associated with development of serious LBP ($p<0.01$) and approached significance for any LBP ($p<0.058$) Neither back lift nor indices of fatigue were associated with development of LBP	
Mostardi *et al.*[177]	Healthy nurses without history of LBP, $n=171$	Isokinetic back lift MVC conducted at baseline Injury reports used to examine incidence of low back injury over 2 years follow-up	9% sustained low back injuries during the follow-up period There was no significant difference in strength at baseline between those who reported low back injury during follow-up and those who did not	

(continued)

Table 4. Continued.

Reference	Participants	Testing	Results	Comments
Cady *et al.*[178]	Healthy fire-fighters without LBP, $n=1652$	Isometric back lift MVC conducted at baseline Incidence of prior low back injuries examined subsequent to baseline measurements—no specific follow-up duration was noted Participants were split into percentiles for 'Most Fit' (84–100%), 'Middle Fit' (17–83%), and 'Least Fit' (0–16%)	7.14% sustained low back injuries in the 'Least Fit' group, 3.19% sustained low back injuries in the 'Middle Fit' group, and 0.77% sustained low back injuries in the 'Most Fit' group	Mean age increased with decreasing fitness levels between the three groups
Mooney *et al.*[179]	Workers without history of LBP in a ship-building firm in the 3 highest Physical Demand Characteristic categories across 32 jobs, $n=152$	Isolated lumbar extension MVC using MEDX 2 year follow-up of low back injury and LBP claims	9% sustained low back injuries during the follow-up period, the majority occurring in the heavy PDC category (64%) Isolated lumbar extension strength was not predictive of low back injuries and only 2 of those participants injured had below normal strength	Age, height. and weight were similar amongst PDC categories and in those injured and uninjured Low back injury rates were significantly higher in heavy and very heavy PDC categories ($p<0.0001$)
Stevenson *et al.*[181]	Spinning operators from DuPont without history of LBP, $n=72$ Spinning operators from DuPont suffering from LBP in previous 2 years, $n=46$ Spinning operators from DuPont suffering from LBP in previous year, $n=31$	EMG recorded bilaterally from lumbar paraspinal muscles at T10 and L3 level 3–4 cm from mid-line during Biering-Sorenson test 2 year follow-up period at 6 month intervals for LBP experiences in previous 6 months	EMG indices of fatigues entered final model and were significantly predictive of LBP ($p=0.035$)	Other factors in final predictive model included age, peak thoracic acceleration, leg strength/endurance, however psychosocial factors were largely absent
Heydari *et al.*[182]	Healthy participants classified as either 'No History of LBP', 'CLBP', or 'Past History of LBP', $n=105$	EMG recorded bilaterally from lumbar paraspinal muscles at L4/L5 level during back lift test maintaining 2/3 MVC for 30 s at baseline and follow-up 2 year follow-up participants were asked to classify themselves as 'worse', 'better', or 'the same'	At follow-up, 76 classified themselves as 'the same', 13 'better', and 16 'worse' EMG indices of fatigue showed greater fatigue was significantly associated with development of first time LBP and with self-classification at follow-up ($p<0.05$)	

The association of deconditioning, specifically of the lumbar extensors in those with CLBP, and as a prospective risk factor for development of LBP, has been demonstrated in numerous studies and with various different methods. Studies conducting specific testing of ILEX evidence that weakness appears localized to the lumbar extensor musculature[96–102] as compared with the quite contrasting evidence utilizing TEX. Imaging studies also demonstrate that deconditioning is consistently found in the ES, MF, and QL of those with CLBP[74,91,107,127]. However, whether this is level- or side-specific is unclear[103,115,117], and it appears that adverse muscle fiber composition is perhaps not present[77]. Finally, excessive fatigability of the lumbar extensors in symptomatic participants has been evidenced more objectively through use of EMG fatigue indices analysis[79,135,154]. Thus, it seems that specific deconditioning of the lumbar extensor musculature may be a common factor in LBP, lending evidence towards the deconditioning hypothesis and to the speculation of other authors regarding its important role[60,183–186]. Further support is shown through prospective studies highlighting that deconditioning may be a risk factor for initial development of LBP[68,83,159,170,174–176,179,181,182].

However, it should be noted that, although a body of research exists to support this hypothesis, there also exists some contrasting evidence to refute it which has been conducted with similarly rigorous methodology[40,77,84,88,176,179]. Indeed we have noted throughout this review the concerns with many of the methodologies employed in much previous research, even amongst the more carefully controlled studies. As such, although the hypothesis is by no means refuted, it still requires further rigorous testing that may be found to either further support or more definitely refute it.

For now, however, we contend that the hypothesis presents a convincing explanation of LBP. The evidence reviewed herein is also supported by other areas of research considered as important to determining causality by the criteria put forth by Hill[161]; criteria such as biological plausibility and experimental reversibility[187,188]. Evidence shows that specifically addressing lumbar extensor deconditioning through ILEX resistance exercise programs in CLBP provides significant reductions in pain and disability[60,97,99,189–200]. There is also evidence suggesting that improvements in ILEX strength correlate with reductions in pain and disability[99,200]. In addition there is evidence that prospectively addressing lumbar extensor deconditioning through ILEX resistance training reduces risk of further low back injury occurring[201–203]. Thus, there is evidence for a relatively consistent prospective and cross-sectional association, biological plausibility through biomechanical modeling studies of lumbar spine stability, experimental reversibility, and also evidence for prospective strengthening to reduce injury risk. These factors combine to offer why deconditioning is perhaps a quite robust account of why LBP is such a wide ranging condition.

One issue that many authors have with this explanation of LBP, however, is very clearly summarized by Crossman *et al.*[77]. They note that, of the studies suggesting the presence of lumbar extensor deconditioning in LBP, 'in none of these studies were any mechanisms offered up to explain how "normal" paraspinal muscle could "dysfunction" to predispose to LBP'. Yet we suggest that the lumbar extensors as an isolated muscle group may exist in a potential state of specific chronic 'disuse', and thus become 'deconditioned' in the first instance independent of physical activity levels due to their anatomy[60]. Indeed this specific state of disuse may stem from the lumbo-pelvic anatomy that is a consequence of our species' evolutionary history; in essence relatively weak lumbar extensors comparable to strong hip extensors[204]. This seems further apparent as most forms of activity and exercise appear to provide little to no conditioning effect[183,186,205–210]. Although, as noted in the introduction, 'disuse' is often considered as a general reduction in physical activity, it seems here that 'disuse' could instead be specifically considered as applicable to the lumbar extensors due to the difficulty in conditioning them, thus leading to their 'deconditioning'. In a sense, specific 'disuse' may lead to specific 'deconditioning' of the lumbar extensors, which may further lead to injury and LBP. However, this is not simply a reduction in general activity levels; it is due to the inability to effectively maintain their condition as a consequence of their anatomy as the hip extensors appear to 'take-over' much of the load bearing[36,41,42].

It should be made clear that it is not the intention of this review to argue for a singular cause of LBP. Although prospective evidence is suggestive of initial deconditioning being a risk factor for development of acute low back injury, LBP and various pain causing mechanisms, and that the majority of acute cases develop into CLBP, this is unlikely to be the only potential causative factor. Many other risk factors have indeed been reported. It is even possible that deconditioning is in fact a result of the impact of pain and other symptoms in some instances[7] and it is likely that both directions of causality could manifest. That being said, however, a body of evidence would appear to implicate specific lumbar extensor deconditioning in LBP, potentially as a primary factor predisposing injury (Figure 2), and thus warrants an addition to the general conceptualization of the 'Disuse/Deconditioning Syndrome'. This also strongly justifies an exercise-based approach designed to effectively recondition the lumbar extensor musculature, regardless of the direction of causality.

That the deconditioning associated with LBP appears for the most part to be mainly localized to the lumbar extensors specifically also warrants that preferably a specific approach towards reconditioning be utilized. Both Helmhout *et al.*[211] and Mayer *et al.*[212] emphasize the issue with many reviews that consider 'exercise' as a single class of treatment without consideration to the variation in exercise approaches that have been applied. Many studies of exercise have also been criticized as lacking an adequate description of the precise exercises used[211,212]. Previous Cochrane reviews have not adequately described, defined, and categorized the 'exercise' studies they have examined, potentially explaining the generally unfavorable conclusions drawn[213,214]. The Cochrane reviews have been specifically criticized for this flaw and their wide-sweeping conclusions[213–216]. Indeed we have also raised this issue of specificity[60,200,217,218]. As noted, research has shown that the lumbar extensors are notoriously difficult to train unless the pelvis is appropriately restrained in order to provide ILEX[183,186,205–210,219]. That we have presented here that deconditioning may be specifically located in the lumbar extensors supports the contention that exercise approaches should specifically address this.

Conclusion

This review has provided a reconsideration of the importance of the deconditioning hypothesis as it relates to the development of, and association with, LBP. Deconditioning of specifically the lumbar extensors appears to be a consistent factor in LBP. However, many of the studies reviewed herein have contained various methodological flaws and so such a conclusion should remain tentative. Future work should seek to further clarify this relationship by acknowledging these and aiming to improve upon the previous research. In addition, the results of this review should perhaps be considered in the design of exercise-based rehabilitation approaches to LBP and also as preventative approaches.

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Supplemental tables

Table A1. Summary of studies testing strength and endurance of the lumbar extensor musculature in LBP.

Reference	Participants	Testing	Results	Comments
Trunk Extension Studies				
Kankaanpaa et al. [37]	Healthy controls without history of LBP, $n = 15$ Middle aged women with CLBP, $n = 20$	Isometric MVC and isometric endurance to failure at 50%MVC during seated (knees 90°) restrained trunk extension	Significantly lower MVC and time endurance time to exhaustion in CLBP (both $p < 0.05$)	Those with previous lumbar surgery were excluded Age, height, body mass and BMI similar between groups
Alaranta et al. [48]	Never any pain, $n = 116$ Pain more than 12 months ago, $n = 46$ Pain during previous 12 month, no disability, $n = 166$ Disabling pain during previous 12 months, $n = 147$	Biering-Sorensen test	Significantly lower endurance time in those with history of LBP ($p < 0.05$)	
McNeil et al. [62]	Healthy controls, $n = 57$ CLBP patients, $n = 40$	Standing trunk extension/ flexion MVC with pelvis restrained at top of iliac crest on superior edge of backboard using a belt across the anterior superior iliac spine, and bilateral restraints upon the iliac crests.	Both extension/flexion were lower in CLBP, however extension was reduced to a significantly greater degree shown by significantly lower extension/flexion ratios ($p < 0.01$)	Participants with sciatica & CLBP had significantly lower extension strength compared to both just CLBP participants and healthy controls ($p < 0.01$) – with the exception of comparison to females with CLBP (*ns*)
Addison & Schultz [63}	Healthy controls, $n = 57$ CLBP patients, $n = 33$	Standing trunk extension/ flexion MVC with pelvis restrained at top of iliac crest on superior edge of backboard using a belt across the anterior superior iliac spine, and bilateral restraints upon the iliac crests.	Both extension/flexion were lower in CLBP, however extension was reduced to a significantly greater degree shown by significantly lower extension/flexion ratios ($p < 0.001$)	No differences between CLBP and an outpatient CLBP group suggesting common physical deficit despite differences in treatment seeking behaviour
Takemasa et al. [64]	Healthy controls without past history of LBP, $n = 126$ CLBP with or without organic lumbar lesions, $n = 123$	Isometric MVC during seated (knees 90°) restrained flexion/ extension	Both flexion/extension were significantly lower in CLBP ($p < 0.05$), however extension was reduced to a significantly greater degree shown by significantly higher flexion/extension ratios in lesion group ($p < 0.01$)	No differences CLBP with or without organic lumbar lesions suggesting common physical deficit despite differences symptoms Age, height, body mass and BMI similar between groups
Handa et al. [65]	Healthy controls without past history of LBP, $n = 60$ CLBP patients, $n = 52$	Isometric MVC during seated (knees 90°) restrained flexion/ extension	Isometric flexion did not significantly differ between groups, isometric extension was significantly lower in CLBP group ($p < 0.05$)	Age, height, body mass and BMI similar between groups

(continued)

Table A1. Continued.

Reference	Participants	Testing	Results	Comments
Suzuki & Endo [67]	Healthy controls without past history of LBP, $n=50$ CLBP patients with or without root impairment, $n=90$	Prone trunk extension MVC and flexion with legs both straight and bent at hips and knees with restraint belts across lower extremities	Both straight leg flexion, and trunk extension were significantly weaker in the CLBP group ($p<0.001$)	Age weight and height similar between groups
Leino et al. [69]	*Baseline participants* Participants with ''Good'' low back status, $n=578$ Participants with ''Intermediate'' low back status, $n=260$ Participants with ''Bad'' low back status, $n=64$ *Follow-up participants* Participants with ''Good'' low back status, $n=239$ Participants with ''Intermediate'' low back status, $n=203$ Participants with ''Bad'' low back status, $n=210$	Standing dynamic (baseline) and isometric (follow-up) trunk extension/flexion MVC with buttock and thighs against a supporting plate and ankles tied by a belt	At baseline dynamic flexion was significantly weaker in those with worse low back status ($p<0.01$) however dynamic extension was significantly weaker only in women ($p<0.05$) At follow-up isometric flexion was significantly weaker in only men with worse low back status ($p=0.01$) however isometric extension was significantly weaker in both men and women ($p<0.05$)	
Bayramoglu et al. [70]	Healthy controls with no history of LBP past 2 years, $n=20$ CLBP patients, $n=25$	Standing trunk extension/flexion MVC with stabilised knees and lower back	Both flexion and extension were significantly weaker in CLBP group ($p<0.05$)	Age and height similar between groups
Nicholaisen & Jorgensen [71]	(Group 1) LBP that made work impossible, $n=17$ (Group 2) LBP but not that hindered work, $n=28$ (Group 3) No history of LBP, $n=32$	Standing trunk extension/flexion MVC a and isometric extension endurance to exhaustion at 60%MVC with stabilised knees and lower back Biering-Sorensen test	No difference in extension/flexion strength between groups. Isometric endurance significantly lower for Group 1 compared to 2 3 in females and males ($p<0.05$) Endurance time significantly lower females for Biering-Sorenson test ($p<0.05$)	Age, weight, height and fat free mass similar between groups except for age being higher in group 1 and weight and fat free mass higher in group 1 for females
Holmstrom et al. [72]	(Group A) Healthy controls with no history of LBP, $n=42$ (Group B) CLBP patients with uncertain or negative clinical assessment, $n=75$ (Group C) CLBP patients with positive clinical assessment, $n=86$	Standing trunk extension/flexion MVC unrestrained lower extremities Biering-Sorensen test	No difference in extension/flexion strength between groups Extension/flexion ratio was significantly lower in Group C compared to A ($p<0.05$) Endurance time significantly lower in both Group C and B compared to A for Biering-Sorenson test ($p<0.01$)	Age, weight and height similar between groups
Salminen et al. [73]	Healthy children, $n=38$ Children with LBP, $n=31$ Children with LBP and sciatica, $n=7$	Biering-Sorensen test Sit up isometric test with knees at 90^0	Both flexion and extension endurance times were significantly lower in LBP groups ($p<0.05$) No difference between LBP and LBP with sciatica was found.	No differences CLBP with or without sciatica suggesting common physical deficit despite differences symptoms Age, sex, school matched between groups

(continued)

Table A1. Continued.

Reference	Participants	Testing	Results	Comments
Hultman et al. [74]	Healthy controls without history of LBP, $n=36$ Patients with intermittent LBP, $n=91$ CLBP patients, $n=21$	Seated isokinetic/isometric trunk extension/flexion with thighs restrained Biering-Sorensen test	All variables, except isokinetic/isometric trunk flexion, were significantly lower in CLBP compared to healthy controls and intermittent LBP patients ($p<0.05$)	Those with previous lumbar surgery were excluded Age, height, body mass and body composition similar between groups
Parkkola et al. [75]	Healthy controls, $n=60$ CLBP patients suitable for active rehabilitation, $n=38$ CLBP patients with serious back problems suitable for moderate rehabilitation only, $n=10$	Standing isometric trunk extension/flexion MVC with chest, thighs and hips restrained	Extension/flexion MVCs showed a gradient between the three groups from higher to lower.	No statistical data reported Incidence of disc degeneration significantly higher in CLBP patients ($p<0.05$) Age, sex, employment and profession matched between groups and BMI similar
Mayer et al. [76]	Healthy controls without history of previous LBP, $n=19$ Postoperative spinal disc surgery patients, $n=46$	Isokinetic trunk extension/flexion peak torque unrestrained lower extremities	Both extension and flexion were significantly lower in the postoperative group ($p<0.05$) with the greatest decrease being in extension strength	There was a significant correlation between trunk extensor strength and muscle density in postoperative patients No information on whether demographic characteristics differed between groups
Crossman et al [77]	Healthy controls without lasting >3 days in previous 12 months, $n=32$ CLBP patients, $n=35$	Standing trunk extension/flexion isometric MVC unrestrained lower extremities Biering-Sorensen test	MVC and endurance time significantly lower in CLBP group ($p<0.05$)	Those with previous lumbar surgery were excluded Age, gender and all anthropometric characteristics similar between groups
Paasuke et al. [78]	Healthy controls, $n=12$ CLBP patients, $n=12$	EMG recorded bilaterally from lumbar paraspinal muscles at L3 level 3 cm from midline during Biering-Sorenson test to failure	Endurance time was significantly lower in the CLBP group ($p<0.05$)	Those with previous lumbar surgery were excluded Age and gender matched between participant groups Age, height, body mass and BMI similar among participant groups
Humphrey et al. [79]	Healthy controls without history of LBP in previous 5 years, $n=175$ CLBP patients, $n=145$ Participants with past history of LBP but no attack within previous 2 years, $n=30$	Back lift MVC	MVC significantly lower in CLBP patients compared to controls ($p<0.01$)	Those with previous lumbar surgery were excluded CLBP group was significantly older and had higher body mass and BMI than controls
Suuden et al. [80]	Healthy controls, $n=20$ CLBP patients, $n=20$	Biering-Sorenson test	Endurance time significantly lower for CLBP patients compared to controls ($p<0.05$)	Those with previous lumbar surgery were excluded Age, height and weight and BMI similar between groups
Lariviere et al. [81]	Healthy controls without history of LBP in previous year, $n=18$ CLBP patients, $n=18$	Dynamic roman chair trunk extensions to failure	Number of repetitions to failure were significantly less in CLBP patients compared to controls ($p<0.001$)	Those with previous lumbar surgery were excluded Age, height, weight and BMI similar between groups
Demoulin et al. [82]	Healthy controls without history of LBP in previous year, $n=10$ CLBP participants, $n=10$	Isometric MVC during seated (knees 90°) restrained trunk extension	Extension strength significantly weaker in CLBP ($p<0.05$)	Those with previous lumbar surgery were excluded

(continued)

Table A1. Continued.

Reference	Participants	Testing	Results	Comments
Balague et al. [86]	Children (10–16yrs) without history of LBP, $n = 79$ Children (10–16yrs) with history of LBP, n 38	Standing isokinetic trunk extension/flexion peak torque unrestrained lower extremities	No significant differences for flexion or extension between groups at any age	
Suter & Lindsay [87]	Healthy controls, $n = 16$ Golfers with CLBP, $n = 25$	Biering-Sorenson test	No significant difference in endurance time between groups	Age, height and weight similar between groups
Da Silva et al. [88]	Healthy controls without history of LBP in previous year, $n = 15$ CLBP patients, $n = 13$	Standing trunk extension, prone trunk extension and back lift MVC	No differences between groups	Those with previous lumbar surgery were excluded Age, height and weight similar between groups
Lariviere et al. [89]	Healthy controls without LBP lasting 1 wk in previous year, $n = 31$ CLBP patients, $n = 27$	Standing trunk extension/ flexion MVC and repetitions to failure (endurance time) with stabilised knees and lower back	No significant difference between health controls and CLBP patients for strength or endurance time Low predicted endurance time was associated with high pain catastrophising in CLBP patients ($p < 0.01$)	Those with previous lumbar surgery were excluded Age, height, weight and BMI similar between groups
Renkawitz et al. [90]	Healthy tennis players without LBP, $n = 36$ Tennis players with CLBP, $n = 48$	Standing isometric trunk extension MVC with shoulders, pelvis and thighs hips restrained	No association between presence of CLBP and trunk extension strength in either univariate or multivariate logistic regressions	
Isolated Lumbar Extension				
Lariviere et al. [40]	Healthy controls without LBP lasting 1 wk in previous year, $n = 18$ CLBP patients, $n = 18$	Isolated lumbar extension MVC and number of repetition to failure at 60%MVC using customised dynamometer	No significant difference between groups	Those with previous lumbar surgery were excluded Age, body mass, height, BMI, body % and physical activity levels were similar between groups
Cassisi et al. [96]	Healthy controls without history of LBP, $n = 12$ CLBP patients, $n = 21$	Isolated lumbar extension MVC using MEDX	Lumbar extension significantly weaker in CLBP ($p = 0.01$)	13 CLBP patients had undergone previous surgery though no effect upon lumbar extension strength was observed Age and height were similar between groups though body mass was greater in CLBP group
Holmes et al. [97]	Healthy geriatric female controls, $n = 20$ CLBP geriatric female patients, $n = 18$	Isolated lumbar extension MVC using MEDX	Lumbar extension significantly weaker in CLBP ($p < 0.05$)	Age, height and weight similar between groups
Robinson et al. [98]	Healthy controls, $n = 12$ CLBP patients (53% having had previous surgery), $n = 16$	Isolated lumbar extension MVC using MEDX was performed and 60%MVC determined at full extension for further EMG analysis during isotonic trial (see table 3)	Absolute load used during isotonic trial was significantly lower in the CLBP group compared with the asymptomatic controls ($p < 0.05$)	10 CLBP patients had undergone previous surgery Age, height and weight similar between groups
Nelson et al. [99]	CLBP patients, $n = 895$	Isolated lumbar extension MVC using MEDX	CLBP baseline data was compared graphically to healthy norms from [53] and shown to considerably weaker.	Patients diagnoses included non-specific CLBP, degenerative disc/arthritic disease, lumbar disc syndrome or spondylolisthesis/ spondylolysis

(continued)

Table A1. Continued.

Reference	Participants	Testing	Results	Comments
Mooney et al. [100]	Strip mine workers (90% reported prior LBP), $n=197$	Isolated lumbar extension MVC using MEDX	Baseline data was compared graphically to healthy norms from [53] and shown to considerably weaker.	
Mooney et al. [101]	Healthy controls, $n=8$ CLBP patients, $n=8$	Isolated lumbar extension MVC using MEDX	CLBP baseline data was compared graphically to both healthy participants in the study and healthy norms from [53] and shown to be considerably weaker.	Patients showed evidence of degenerative disc disease
Boyce et al. [102]	Small manufacturing plant workers (53% reported LBP), $n=20$	Isolated lumbar extension MVC using MEDX	Baseline data was compared graphically to healthy norms from [53] and shown to considerably weaker.	

Table A2. Summary of imaging and histochemical studies of the lumbar extensor musculature in LBP.

Reference	Participants	Testing	Results	Comments
Imaging Studies				
Hultman et al. [74]	Healthy controls without history of LBP, $n=24$ Patients with intermittent LBP, $n=40$ CLBP patients, $n=21$	CSA and density of erector spinae using CT at L3 level	Muscle density was significantly lower in CLBP patients compared to both other groups ($p<0.05$) CSA did not significantly differ between groups	Those with previous lumbar surgery were excluded Age, height, body mass and body composition similar between groups
Parkkola et al. [75]	Healthy controls, $n=60$ CLBP patients suitable for active rehabilitation, $n=38$ CLBP patients with serious back problems suitable for moderate rehabilitation only, $n=10$	CSA, fat content and grading status graded using 4 classification system of psoas and back muscles (erector spinae and multifidus) using MRI at L4/L5 level	CSA was significantly lower in both CLBP groups compared with controls ($p<0.001$) Back muscle status showed a gradient between the three groups from better to worse. It was significantly worse in severe CLBP patients compared with mild CLBP patients ($p<0.05$) and healthy controls ($p<0.001$), and was significantly worse in mild CLBP patients compared with controls also ($p<0.05$) Psoas muscles did not differ between groups	Incidence of disc degeneration significantly higher in CLBP patients ($p<0.05$) Age, sex, employment and profession matched between groups
Mayer et al. [76]	Healthy controls without history of previous LBP, $n=19$ Postoperative spinal disc surgery patients, $n=46$	CSA and muscle density of psoas, erector spinae, rectus abdominus and obliques using CT at L3	Non-significant trends towards reduced CSA in psoas and erector spinae were found in the postoperative group Muscle density of psoas and erector spinae was significantly lower in the postoperative group ($p<0.001$)	There was a significant correlation between trunk extensor strength and muscle density in postoperative patients No information on whether demographic characteristics differed between groups

(continued)

Table A2. Continued.

Reference	Participants	Testing	Results	Comments
Kamaz et al. [91]	Healthy controls without LBP or leg pain, $n = 34$ CLBP patients, $n = 36$	CSA of total paraspinal, multifidus, quadratus lumborum, psoas and gluteus maximus muscles using CT at L4 upper and lower plates	CSA was significantly reduced in only paraspinal and multifidus at the lower plate in CLBP ($p < 0.01$) CSA was significantly reduced in only multifidus, psoas and quadratus lumborum at the upper plate in CLBP ($p = 0.05$) No significant differences between CSA of gluteus maximus	Those with previous lumbar surgery were excluded Age and BMI similar in both groups.
Sihvonen et al. [94]	LBP patients who underwent surgery for lumbar spinal stenosis and/or disc herniation 2–6 years prior with good recovery, $n = 14$ LBP patients who underwent surgery for lumbar spinal stenosis and/or disc herniation 2–6 years prior regarded as post-operatively failed, $n = 21$	Paraspinal muscle density at L4-L5 level using CT	Muscle density was significantly greater in the group with good recovery compared with the post-operatively failed group ($p < 0.01$)	Lumbar spinal stenosis and/or disc herniation confirmed by CT Age similar between groups
Mooney et al. [101]	Healthy controls, $n = 8$ CLBP patients, $n = 8$	Fatty infiltration and CSA of lumbar paraspinal musculature using MRI from L3 endplate to lower endplate of L5 and graded using 4 classification system	CLBP patients showed evidence of fatty infiltration compared with controls 5/8 showing severe All patients showed greater fatty infiltration of paraspinal muscles compared with any other lumbar muscles No difference in CSA between groups	No statistical data reported Patients showed evidence of degenerative disc disease
Hides et al. [103]	Healthy controls, $n = 51$ First episode acute LBP patients, $n = 26$	CSA of multifidus on left and right sides using real-time ultrasound at L2, L3, L4, L5 and S1	Asymmetry was significantly greater corresponding to level of symptoms in LBP patients compared with normal participant between-side differences ($p < 0.001$)	Only comparisons of between side differences were reported between LBP patients and normal participants. Manual extraction of data on CSA from figure 2 in ref [96] suggests that average CSA of asymptomatic side in LBP patients did not differ significantly from healthy participant's largest side. Age, height and weight similar between groups
Mannion et al. [106]	CLBP patients, $n = 59$	CSA of erector spinae, quadratus lumborum and psoas using MRI at L3/L4 and L4/L5 levels	CSA showed association with lean body mass and age, but no association with symptom duration	No healthy control group for comparisons Those with previous lumbar surgery were excluded

(continued)

Table A2. Continued.

Reference	Participants	Testing	Results	Comments
Mengiardi et al. [107]	Healthy controls without history of LBP in previous 2 years, $n=25$ CLBP patients, $n=25$	CSA of multifidus and longissimus fat content and semi-quantitative grading using 5 classification system using MRI at L4-L5 level	CLBP patients showed significantly greater fat content in the multifidus ($p<0.05$) No difference found using semi-quantitative system	Those with previous lumbar surgery were excluded Age, sex and BMI matched between participant groups
Cooper et al. [111]	Recent onset LBP patients (symptoms less than 18 months), $n=43$ CLBP patients (symptoms more than 18 months), $n=44$	CSA of paraspinal and psoas muscles using CT at L4 normalised to L4 bone CSA	Normalised paraspinal and psoas CSAs significantly reduced in CLBP compared to recent onset group ($p<0.05$)	All participants technically chronic as defined by Frymoyer [108] Lumbar surgery in preceding 18 months were excluded, though most CLBP patients included ($n=31$) had undergone prior surgery CLBP participants also significantly older
Bouche et al. [112]	Post-discectomy patients pain free, $n=18$ Post-discectomy patients with LBP, $n=18$	Muscle CSA and fat CSA of total paraspinal, erector spinae, multifidus and psoas + iliac muscle using CT at L3, L4, and L5 normalised to L3 bone CSA	Muscle CSA of erector spinae and multifidus significantly smaller in pain patients ($p<0.05$) Fat CSA significantly greater in psoas of pain patients ($p<0.05$)	Level of operation was not found to be a significant factor and so suggests a general deconditioning of the lumbar musculature independent of surgery Age and BMI similar between groups
Danneels et al. [127]	Healthy controls without history of previous LBP, $n=23$ CLBP patients, $n=32$	Total CSA and muscle CSA of total paraspinal, erector spinae, multifidus and psoas muscles using CT at upper L3, and upper and lower L4 normalised	Total CSA of paraspinal and multifidus muscles significantly smaller at lower L4 in CLBP ($p<0.05$) No significant difference for erector spinae or psoas	Those with previous lumbar surgery were excluded in addition to those who had participated in training for the lower back muscles in the previous 3 months Age, height, weight and activity similar between groups
Alaranta et al. [113]	CLBP patients, $n=39$	Fat content of lumbar paraspinal musculature using CT at three lowest levels and 4 level classification system	Fat content was moderately positively associated with disability score on Oswestry index ($p<0.05$) but not with age, sex, body mass, BMI, degree of disc degeneration, or facet joint osteoarthritis	Those with previous spinal fusion surgery were excluded however 16 patients had undergone previous surgery for lumbar disc herniation > 1 year prior.
Kader et al [114]	CLBP patients, $n=75$	Atrophy of the multifidus compared with normal results from [106] using MRI and 3 level classification system	80% of participants showed moderate of severe multifidus atrophy	Those with previous lumbar surgery were excluded Significant association between multifidus atrophy and leg pain ($p<0.01$)

(continued)

Table A2. Continued.

Reference	Participants	Testing	Results	Comments
Barker et al. [115]	CLBP patients with unilateral pain, $n=50$	CSA of left and right multifidus and psoas muscles using MRI at level of symptoms and one level above and below	CSA of both multifidus and psoas significantly smaller on symptomatic side at all levels ($p<0.05$)	Those with previous lumbar surgery were excluded Multifdus atrophy consistently relatively greater than psoas atrophy at all levels Significant association between psoas atrophy and pain, nerve root compression and symptom duration. Significant association between multifidus atrophy and symptom duration
Kjaer et al. [116]	Adults aged 40 years, $n=409$ (85% reporting LBP ever, 70% reporting LBP in previous year) Adolescents aged 13 years, $n=439$ (41% reporting LBP ever, 22% reporting LBP in previous year)	Fat content of multifidus using MRI at 3 lower lumbar levels using 3 level classification system	Association between fat content of multifidus for LBP ever (Odds Ratio $=7.2$) and LBP in previous year (Odds Ratio $=3.6$) in adults. No association between fat content of multifidus in adolescents	Associations increased when controlling for effect moderators including gender, BMI, physical workload, leisure and sports activities.
Hyun et al. [117]	Healthy controls without lumbosacral radiculopathy or disc herniation, $n=19$ LBP patients with unilateral lumbosacral radiculopathy, $n=14$ LBP patients with disc herniation but no lumbosacral radiculopathy, $n=25$	Total CSA and muscle CSA of mulitifidus using MRI at L3/L4, L4/L5, and L5/S1	Total CSA, muscle CSA and ratio of the two were significantly reduced in both LBP groups involved sides compared to controls at most levels ($p<0.05$) and ratio at L3/L4 ($p<0.05$) No difference between LBP groups for total CSA, muscle CSA and ratio of the two Ratio of involved side CSA to uninvolved side CSA was significantly different in radiculopathy patients compared to both controls and the other LBP patients ($p<0.01$ to 0.05)	Those with previous lumbar surgery were excluded from LBP control group
Kalichman et al. [118]	Healthy controls without LBP in previous year, $n=150$ Patients who had suffered from LBP of at least 1 month within previous year, $n=37$	Density of erector spinae and multifidus muscles using CT at L3, L4, and L5	Muscle density was not associated with LBP Reduced muscle density was significantly associated with presence of facet joint osteoarthritis, spondylolisthesis and disc narrowing ($p<0.05$)	

(continued)

Table A2. Continued.

Reference	Participants	Testing	Results	Comments
Hicks et al. [119]	Controls aged 70–79 years without LBP in previous year, $n=861$ Patients with mild LBP in previous 12 months, $n=244$ Patients with moderate LBP in previous 12 months, $n=299$ Patients with severe/ extreme LBP in previous 12 months, $n=111$	Total CSA and density of paraspinal, and lateral abdominal muscles using CT at L4-L5 level	Both non-adjusted and adjusted means for muscle density showed significant associations with the presence and severity of LBP for the paraspinal muscles ($p \leq 0.0001$), and lateral abdominals ($p<0.05$).	
Kang et al.[128]	CLBP patients with lumbar degenerative kyphosis undergoing corrective surgery, $n=54$ CLBP control patients, $n=54$	CSA and muscle to disc CSA ratio of psoas, erector spinae and multifidus was assessed at L4/L5 level and fatty infiltration of psoas, erector spinae and multifidus assessed at L3/L4 using three grade classification using MRI	CSA and muscle to disc CSA ratios for all muscles were significantly lower in the lumbar degenerative kyphosis group compared with controls ($p<0.001$) with regression analysis showing multifidus wasting to be most strongly associated ($p<0.001$) Severe fatty infiltration was significantly more common in lumbar degenerative kyphosis compared to CLBP controls ($p<0.05$)	No healthy control group for comparisons Those with previous lumbar surgery were excluded from CLBP control group Age and sex matched between groups and symptom durations were similar Body mass and BMI was significantly higher in CLBP controls No difference in degenerative changes (degenerative disc disease, herniation's, stenosis or spondylolithesis) between groups
Histochemical Studies				
Crossman et al [77]	Healthy controls without LBP lasting > 3 days in previous 12 months, $n=32$ CLBP patients, $n=35$	Percutaneous biopsy of paraspinal muscle (specific location not noted) for fibre CSAs and fibre typing.	No significant differences between groups for any fibre histochemical comparisons	Those with previous lumbar surgery were excluded Age, gender and all anthropometric characteristics similar between groups
Weber et al. [92]	LBP patients undergoing posterior surgery, $n=61$ (posterior surgery for persistent pain Op1 $n=43$, posterior surgery for removal of internal fixation Op2 $n=32$)	Biopsy of multifidus at L3, L4, L5 or S1 level for fibre diameter, fibre typing and pathological changes	Pathological changes were common in biopsy specimens from Op1 Type II atrophy was associated with age and severity of pain in biopsy specimens from Op1 Patients undergoing Op2 showed significantly greater pathological changes compared with biopsy specimens from Op 1 ($p=0.05$) Biopsy specimens were taken from 14 patients at the same level in both Op1 and Op2 with 70% of normal biopsies at Op1 showing alterations at Op2	No healthy control group for comparisons Muscular alterations were present in patients undergoing Op1 however surgery may have caused further alterations as presence of changes were increased in Op2

(continued)

Table A2. Continued.

Reference	Participants	Testing	Results	Comments
Rantanen et al [93]	Patients from ref [123] who underwent surgery for lumbar disc herniation 5 years prior, $n=18$	Biopsy of multifidus taken 1 cm laterally from spinous process of the level immediately below the previously herniated disc (L4/L5 and/or L5/S1) for fibre narrow diameter, fibre typing, atrophy/hypertrophy and pathological changes	No changes in fibre type distribution, atrophy/hypertrophy factors were noted compared with baseline Type I fibre size significantly increases	Level of herniation and thus biopsy did not influence results Patients with both 'positive' and 'negative' outcomes from original surgery were compared showing decreased pathological changes in 'positive' group compared with their persistence in 'negative' group
Sihvonen et al. [94]	LBP patients who underwent surgery for lumbar spinal stenosis and/or disc herniation 2–6 years prior with good recovery, $n=14$ LBP patients who underwent surgery for lumbar spinal stenosis and/or disc herniation 2–6 years prior regarded as post-operatively failed, $n=21$	Biopsy of paraspinal muscle taken from site of abnormal myelogram finding for fibre atrophy	Local denervation atrophy observed in all but one post operatively failed patients	Lumbar spinal stenosis and/or disc herniation confirmed as absent by CT Age similar between groups No statistical data reported
Mannion et al. [106]	CLBP patients, $n=59$	Biopsy of belly of lateral tract of left erector spinae at L3/L4 level for fibre CSA, fibre typing and pathological changes	Symptom duration was a strong predictor of both fibre type changes towards a more type IIx phenotype Pathological changes were common and significantly associated with age and showed a trend to association with symptom duration	No healthy control group for comparisons Those with previous lumbar surgery were excluded
Ford et al. [109]	Patients undergoing surgery for lumbar disc herniation reporting LBP duration between 3 and 52 weeks, $n=18$	Biopsy of erector spinae (sacrospinalis) 1 cm lateral to tip of spinous process and multifidus 1 cm from inferior border of lamina at L5 level for fibre typing, fibre narrow diameter and pathological changes	No differences between left and right sides Pathological changes were common but varied and not impacted by side of herniation	No healthy control group for comparisons Side of herniation did not affect results
Zhu et al. [120]	Patients undergoing surgery for lumbar disc herniation, $n=22$	Biopsy of erector spinae from side and level of herniation 1 cm lateral to top of spinous process for fibre typing, atrophy/hypertrophy and pathological changes	Proportion of fibres types for type I, type IIa and type IIb were 68%, 10.6% and 21.4% respectively Type II atrophy was common with type IIb most frequent and severe 18 patients showed evidence of pathological changes	No healthy control group for comparisons

(continued)

Table A2. Continued.

Reference	Participants	Testing	Results	Comments
Mannion et al. [121]	Healthy controls without history of LBP requiring time of work or doctors attention, $n = 29$ CLBP patients undergoing posterior surgery, $n = 31$ (First time operation $n = 22$, patients undergoing second operation $n = 9$)	Biopsy of belly of lateral tract of left erector spinae at L3 level for fibre narrow diameter, fibre typing and pathological changes	Smaller proportion of type I and greater proportion of type IIb fibres as both % and % fibre type area were found in CLBP patients compared to healthy controls ($p < 0.05$) Pathological changes did not differ between groups	Age, sex and body mass matched between participant groups
Fidler et al. [122]	Patients with LBP, $n = 17$ Cadavers within 24 hours of death, $n = 3$	Biopsy of multifidus from separated muscle cut transversely, taken during operation	Grouping of slow fibres appeared in addition to reduced CSA of fast fibres in LBP	No details on nature of operation No statistical data reported
Mattila et al [123]	Patients undergoing first time surgery for lumbar disc herniation, $n = 41$ Control participants without history of LBP undergoing autopsy within 48 hours of death, $n = 12$	Biopsy of multifidus taken during operation or autopsy at L4/L5 and L5/S1 levels for fibre narrow diameter, fibre typing, atrophy/hypertrophy and pathological changes	Relative numbers of type I and type II fibres did not correlate with age nor differ significantly between groups Pathological changes were significantly more frequent in patients compared to controls ($p < 0.01$)	Biopsy taken from deltoid to rule out systemic congenital myopathy Level of herniation and thus biopsy did not influence results
Zhao et al. [124]	LBP patients undergoing first time surgery for lumbar disc herniation, $n = 19$	Biopsy of multifidus taken during operation from transversospinal corner on both left and right sides at the level of herniation (L4/L5 or L5/S1) for fibre CSA, fibre narrow diameter, fibre typing and pathological changes	CSAs and diameters of both type I and type II fibres were significantly smaller on the side of herniation ($p < 0.05$) Strength factor (%fibre type x fibre CSA) of type II fibres was also lower on side of herniation ($p < 0.05$) Pathological changes were present in both sides but more severe on the side of herniation	No healthy control group for comparisons Location of pain symptoms was associated with muscle alterations
Bajek et al. [125]	Patients undergoing surgery for lumbar disc herniation, $n = 76$ Control participants without history of neuromuscular disease undergoing autopsy within 48 hours of sudden death, $n = 41$	Biopsy of multifidus on side of herniation and at level of herniation in patients (L3/L4, L4/L5, or L5/S1) and L4/L5 level in controls 1 cm lateral from midline deeper than the aponeurosis of erector spinae for fibre typing and fibre diameter	Greater proportion of type I and smaller proportion of type IIa type IIb fibres in patients compared with controls in males only ($p < 0.05$) Fibre diameter in type I fibres was significantly greater in paitents compared to controls ($p < 0.05$) and for type IIa and type IIb was significantly greater than controls for males only ($p < 0.05$)	Age was similar between groups

(continued)

Table A2. Continued.

Reference	Participants	Testing	Results	Comments
Yoshihara et al. [126]	LBP patients undergoing first time surgery for lumbar disc herniation, $n=29$	Biopsy of multifidus taken during operation immediately after start of surgery dissected from L4 and L5 muscle bands on both sides for fibre typing, fibre size and pathological changes	Fibre size of type 2 fibres was significantly smaller than type I at all biopsy sites Fibre size did not differ between sides at L4 for type I or type II fibres but fibre size was significantly smaller at L5 on side of herniation for both type I and type II fibres ($p<0.01$) No difference in fibre type proportions Pathological changes were present at all biopsy sites but only significantly different between sides, with greater frequency on side of herniation at L5	No healthy control group for comparisons No difference between level of biopsy

Table A3. Summary of studies testing fatigability with EMG of the lumbar extensor musculature in LBP.

Reference	Participants	Testing	Results	Comments
Kankaanpaa et al. [37]	Healthy controls without history of LBP, $n=15$ Middle aged women with CLBP, $n=20$	EMG recorded bilaterally from gluteus muscles and lumbar paraspinal muscles at L3/L4 and L5/S1 levels 2 cm laterally from midline of spinous process during isometric MVC and isometric endurance to failure at 50%MVC during seated (knees 90°) restrained trunk extension	Neither EMG amplitude or fatigue indices data differed between groups for the paraspinal muscles	Those with previous lumbar surgery were excluded Age, height, body mass and BMI similar between groups
Lariviere et al. [40]	Healthy controls without LBP lasting 1 wk in previous year, $n=18$ CLBP patients, $n=18$	EMG recorded bilaterally from gluteus maximus, biceps femoris and vastus medialis muscles and lumbar paraspinal muscles at L4, L3, L1, and T10 levels during isolated lumbar extension MVC and repetitions to failure at 60%MVC using customised dynamometer	None of the EMG fatigue indices data differed between groups for the paraspinal muscles	Those with previous lumbar surgery were excluded Age, body mass, height, BMI, body % and physical activity levels were similar between groups
Crossman et al [77]	Healthy controls without lasting >3 days in previous 12 months, $n=32$ CLBP patients, $n=35$	EMG recorded biltaerelly from lumbar paraspinal muscles at L4-L5 level during standing isometric trunk extension for 60 seconds at 60%MVC and during the Biering-Sorensen test	EMG fatigue indices were similar between groups for the Biering-Sorenson test and also the 60%MVC test	Those with previous lumbar surgery were excluded Age, gender and all anthropometric characteristics similar between groups

(continued)

Table A3. Continued.

Reference	Participants	Testing	Results	Comments
Paasuke et al. [78]	Healthy controls without history of LBP or LBP in previous year, $n=12$ CLBP patients, $n=12$	EMG recorded bilaterally from lumbar paraspinal muscles at L3 level 3 cm from midline during Biering-Sorenson test to failure	EMG indices of fatigue showed significantly greater fatigue in the CLBP group compared to controls ($p<0.05$)	Those with previous lumbar surgery were excluded Age and gender matched between participant groups Age, height, body mass and BMI were similar between participant groups
Humphrey et al. [79]	Healthy controls without history of LBP in previous 5 years, $n=175$ CLBP patients, $n=145$ Participants with past history of LBP but no attack within previous 2 years, $n=30$	EMG recorded bilaterally from lumbar paraspinal muscles at L4/L5 during a back lift test with 66.66%MVC for 30 seconds	EMG indices of fatigue showed significantly greater fatigue in the CLBP compared to controls ($p<0.05$) Logistic regression showed high sensitivity (0.65) and specificity (0.75) in classifying CLBP patients Past history participants could not be adequately discriminated from either group	Those with previous lumbar surgery were excluded CLBP group was significantly older and had higher body mass and BMI than controls
Suuden et al. [80]	Healthy controls, $n=20$ CLBP patients, $n=20$	EMG recorded bilaterally from lumbar paraspinal muscles at L3 3 cm from midline during Biering-Sorenson test to failure	No significant differences in EMG indices of fatigue between groups	Those with previous lumbar surgery were excluded Age, height and weight and BMI similar between groups
Lariviere et al. [81]	Healthy controls without history of LBP in previous year, $n=18$ CLBP patients, $n=18$	EMG recorded bilaterally from gluteus maximus, biceps femoris and lumbar paraspinal muscles at L4, L3, L1, and T10 levels during dynamic roman chair trunk extensions to failure	No significant differences in EMG indices of fatigue between groups	Those with previous lumbar surgery were excluded Age, height, weight and BMI similar between groups
Suter & Lindsay [87]	Healthy controls without history of LBP, $n=16$ Golfers with CLBP, $n=25$	EMG recorded bilaterally from lumbar paraspinal muscles at T12 and L4-L5 level 3 cm from midline during Biering-Sorenson test to failure	No significant difference in EMG fatigue indices between groups	Age, height and weight similar between groups
Da Silva et al. [88]	Healthy controls without history of LBP in previous year, $n=15$ CLBP patients, $n=13$	EMG recorded bilaterally from lumbar paraspinal muscles at T10, L1, L3, and L5 levels during standing trunk extension and back lift at 50%MVC for 60 seconds, and during Biering-Sorenson test for 60 seconds	No difference in EMG fatigue indices between groups	Those with previous lumbar surgery were excluded Age, height and weight similar between groups

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Table A3. Continued.

Reference	Participants	Testing	Results	Comments
Lariviere et al. [89]	Healthy controls without LBP lasting 1 wk in previous year, $n=31$ CLBP patients, $n=27$	EMG recorded bilaterally from gluteus maximus, biceps femoris and lumbar paraspinal muscles at L5, L3, L1, and T10 levels during standing trunk extension/flexion MVC and repetitions to failure (endurance time) with stabilised knees and lower back	EMG indices of fatigue showed significantly greater fatigue in CLBP patients with high catastrophising compared with CLBP patients with low catastrophising ($p<0.01$)	Those with previous lumbar surgery were excluded Age, height, weight and BMI similar between groups
Robinson et al. [98]	Healthy controls never treated for LBP and without LBP in previous year, $n=12$ CLBP patients (53% having had previous surgery), $n=16$	EMG recorded bilaterally from lumbar paraspinal muscles at L1-L2 level during isolated lumbar extension at 60%MVC in full extension for 12–13 repetitions	EMG amplitude in millivolts decreased across repetitions in asymptomatic participants compared with a significantly flatter curve in the CLBP group ($p<0.05$)	Age, height and weight similar between groups
Roy et al. [135]	Healthy controls, $n=12$ CLBP patients, $n=12$	EMG recorded bilaterally from lumbar paraspinal muscles at L1, L2 and L5 levels during standing isometric trunk extension for 60 seconds at 40%MVC, 60%MVC and 80%MVC	Discriminant analysis of EMG fatigue indices successfully classified 92% controls, 82% CLBP at 40%MVC, 67% controls, 75% CLBP at 60%MVC and 84% controls, 91% CLBP at 80% MVC	Those with previous lumbar surgery were excluded Age, height and weight similar between groups
Roy et al. [151]	Healthy controls without history of LBP, $n=42$ CLBP patients (43% having had previous surgery), $n=28$	EMG recorded bilaterally from lumbar paraspinal muscles at L1, L2 and L5 levels during standing isometric trunk extension for 30 seconds at 40%MVC and 80%MVC	Discriminant analysis of EMG fatigue indices successfully classified 85% CLBP patients and 86% healthy controls	CLBP patients heterogeneous with respect to symptoms and history (75% had disc herniation and 43% had undergone previous surgery)
Mayer et al. [154]	Healthy controls, $n=11$ CLBP patients, $n=10$	EMG recorded bilaterally from lumbar paraspinal muscles at L3 level 3 cm from midline during 10 isometric trunk extension holds on a roman chair lasting 15 seconds each and with 10 seconds rest between each hold	EMG indices of fatigue showed significantly greater fatigue in the CLBP group compared to controls ($p<0.01$)	Those with previous lumbar surgery at level of EMG placement were excluded Age and torso weight similar between groups
Peach & McGill [155]	Healthy controls without history of LBP in previous 2 years, $n=18$ CLBP patients, $n=21$	EMG recorded from lumbar paraspinal muscles at T9 level 5 cm from midline, L3 level 3 cm from midline, and L5 level 1–2 cm from midline respectively during semi-standing isometric trunk extension for 30 seconds at 60%MVC and then after a 60 second rest during a further 10 second extension at 60%MVC	EMG indices of fatigue showed significantly greater fatigue in the CLBP compared to controls ($p<0.05$) Discriminant analysis of EMG fatigue indices successfully classified 100% controls and 93.75% CLBP patients Logistic regression was equally powerful using two parameters with concordance of 92.4%	Age, height and weight similar between groups

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Table A3. Continued.

Reference	Participants	Testing	Results	Comments
Roy et al. [156]	Varsity rowers without LBP, $n = 17$ Varsity rowers with LBP in past year, $n = 6$	EMG recorded bilaterally from lumbar paraspinal muscles at L1, L2 and L5 levels during standing isometric trunk extension for 30 seconds at 80%MVC and then after a 60 second rest during a further 5 second extension at 80%MVC	Discriminant analysis of EMG fatigue indices successfully classified 93% controls and 100% LBP participants	Age, height and weight similar between groups
Biedermann et al. [157]	Healthy controls without history of LBP, $n = 22$ CLBP patients, $n = 27$	EMG recorded bilaterally from lumbar paraspinal muscles at L2-L3 and L4-L5 levels during standing with a 11.6 pound dumbbell held in outstretched arms for 45 seconds followed by a 5 minute recovery and the repetition of the 45 second trial – all adjusted for arm length differences	CLBP patients were classified into 'avoiders' or 'confronters' Discriminant analysis of EMG fatigue indices successfully classified 88.9% 'avoiders', 66.7% 'confronters' and 59.1% controls	Age, height, weight and arm length similar between groups Continuum of fatigue seen between avoiders $>$ confronters $>$ controls, however pain duration differed significantly between avoiders and confronters (8.57 ± 6.22 years and 1.60 ± 0.76 years respectively)
Klein et al. [158]	Varsity rowers without LBP, $n = 17$ Varsity rowers with LBP in past year, $n = 8$	EMG recorded bilaterally from lumbar paraspinal muscles at L1, L2 and L5 levels during standing isometric trunk extension for 30 seconds at 80%MVC and then further 10 second extensions at 80%MVC at 1 minute, 2 minutes, 5 minutes, 10 minutes and 15 minutes into recovery	Discriminant analysis of EMG fatigue indices showed most successful classification at 1 and 2 minute recovery, classifying for 1 and 2 minutes respectively 88% and 100% of LBP participants and 100% and 88% of controls	Age, height and weight similar between groups
Mannion et al. [159]	Healthy controls without history of LBP, $n = 10$ LBP patients, $n = 12$	EMG recorded bilaterally from lumbar paraspinal muscles at T10 and L3 level 3–4 cm from midline during Biering-Sorenson test for 60 seconds	MFS was greater in LBP group indicating greater fatigue but just failed to achieve significance ($p = 0.10$)	Age, height and weight similar between groups Mean values for MFS were similar to those in prospective study which did achieve significance in predicting first time LBP

Table A4. Summary of prospective studies of lumbar extensor musculature deconditioning in LBP.

Reference	Participants	Testing	Results	Comments
Biering-Sorenson [68]	Men aged between 30, 40, 50, and 60 years old, $n = 449$ Women aged between 30, 40, 50, and 60 years old	Biering-Sorensen test conducted at baseline 1 year follow-up with questionnaire concerning first time occurrence, recurrence or persistence of LBP	First time occurrence was significantly associated with low endurance time	
Leino et al. [69]	*Baseline participants* Participants with ''Good'' low back status, $n = 578$ Participants with ''Intermediate'' low back status, $n = 260$ Participants with ''Bad'' low back status, $n = 64$ *Follow-up participants* Participants with ''Good'' low back status, $n = 239$ Participants with ''Intermediate'' low back status, $n = 203$ Participants with ''Bad'' low back status, $n = 210$	Standing dynamic trunk extension/flexion maximum repetitions performed over 30 seconds with buttock and thighs against a supporting plate and ankles tied by a belt conducted at baseline Standing isometric trunk extension/flexion MVC with buttock and thighs against a supporting plate and ankles tied by a belt conducted at 10 year follow-up in addition to questionnaire and assessment of low back symptoms and status	Trunk strength was not predictive of low back symptoms or status at follow up.	
Luoto et al. [83]	Healthy participants without history of LBP in previous year at baseline, $n = 167$	Biering-Sorensen test and questionnaire regarding previous and present LBP conducted at baseline 75% of participants were available for follow-up at 1 year with the same questionnaire, $n = 126$	Endurance time was significantly associated with first time occurrence of LBP when adjusted for age, sex and occupation ($p < 0.05$) Endurance time broken into tertiles (poor, medium, good) showed a non- linear dose-response relationship with first time occurrence of LBP ($p < 0.04$) Relative odds ratio compared to 'good' for 'medium' and 'poor' were1.4 (95% CI 0.4–4.2) and 3.4 (95% CI 1.2–10.0) respectively	

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Table A4. Continued.

Reference	Participants	Testing	Results	Comments
Gibbons et al. [84]	Healthy participants without history of LBP in previous year at baseline, $n=43$	Isokinetic back lift MVC, psychophysical back lift test, Biering-Sorensen test, CSA, proton-density weighted signal, and T2-weighted signal of erector spinae, quadratus lumborum, psoas major and total paraspinal muscle using MRI, and interview regarding previous and present LBP conducted at baseline Interviews regarding LBP were conducted at 1 year follow-up	Neither back lift, psychophysical back lift or endurance time differed between those with and without LBP at follow-up, nor where they associated with frequency of LBP at follow-up Neither CSA, proton-density weighted signal, or T2-weighted signal differed between those with and without LBP at follow-up, however, total paraspinal CSA, and proton-density weighted signal andT2-weighted signal of erector spinae, quadratus lumborum, psoas major were significantly associated with frequency of LBP at follow-up ($p<0.05$)	
Mannion et al. [159]	Healthy nurses without history of LBP, $n=200$	EMG recorded bilaterally from lumbar paraspinal muscles at T10 and L3 level 3–4 cm from midline during Biering-Sorenson test and maintenance of 80%MVC for 28 seconds at baseline Postal questionnaire regarding LBP conducted at 1 year follow-up	13% developed serious first time LBP during the follow-up period EMG indices of fatigue during Biering-Sorenson showed greater fatigue was significantly associated with development of first time LBP at follow-up ($p<0.05$) however endurance time was not associated with first time LBP	
Rissanen et al. [163]	Participants from the Mini-Finland Health Survey, $n=535$	Dynamic trunk extension/flexion maximum repetitions performed over 30 seconds with buttock and thighs against a supporting plate and ankles tied by a belt conducted at baseline Average 12 year follow-up to time until retirement due to work disablement, death or end of observation period for primary diagnosis as cause of work disability	At follow-up of 56 incident cases 15 were due to back disorders Adjusted relative risks in multiple models showed trunk extension performance significantly predicted back disorder disability risk ($p=0.04$–0.002)	
Newton et al. [167]	Healthy participants without history of LBP, $n=70$	Isokinetic trunk extension, flexion, rotation, and back lift MVC and psychophysical lift conducted at baseline 1 year follow-up with questionnaire concerning first time occurrence, recurrence or persistence of LBP	23% developed LBP during the follow-up period, yet at least 6 months after initial assessment in all cases None of the isokinetic measures differed between those who did and those who did not develop LBP	Those with previous lumbar surgery were excluded

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Table A4. Continued.

Reference	Participants	Testing	Results	Comments
Reimer et al. [168]	Healthy prospective order selector employees for 1989, $n=122$ Healthy prospective order selector employees for 1990, $n=122$ Healthy prospective order selector employees for 1991, $n=122$	Dynamic lift capacity, isokinetic trunk extension, flexion, rotation, and back lift MVC and psychophysical lift conducted at baseline to determine placement in employment as an order selector in a warehouse grocery distributor 2 year follow-up with questionnaire concerning first time occurrence, recurrence or persistence of LBP	After implementation of prospective evaluation for employment placement in 1989, incidence of low back injuries were significantly reduced by 32% in 1990 and 41% in 1991 ($p<0.001$)	
Batti'e et al. [169]	Employees working for a large aircraft manufacturer ($n=497$ reporting LBP in previous 10 years), $n=2178$	Isometric MVC for torso, arm and leg lift was conducted at baseline 4 year follow-up conducted for claims related to low back injuries or LBP	Participants with higher MVC for arm, leg and torso lift were at higher risk for LBP and low back injury ($p=0.01$, 0.03, and 0.26 respectively). When adjusted for age and sex however no association was present.	Due to an injury rate of 0.6% during torso lift testing it was discontinued. $n=495$ participants completed torso lift testing, $n=2158$ completed arm lift testing, and $n=2102$ completed leg lift testing
Lee et al. [170]	Healthy student participants without history of LBP, $n=67$	Isokinetic trunk extension, flexion, and rotation MVC conducted at baseline. 5 year follow-up concerning LBP incidence	27% developed first time LBP during the follow-up period Ratio of extension/flexion strength at baseline was significantly lower in participants who developed first time LBP, ($p<0.05$)	Age, height, weight and smoking habits similar between groups
Kujala et al. [171]	Healthy participants without history of LBP, $n=262$	Standing isometric trunk extension/flexion MVC was conducted at baseline 5 year follow-up with questionnaire was conducted regarding type, frequency, severity and functional limitations of LBP	47% developed first time LBP during the follow-up period, 11% of these reporting it as being of monthly frequency, 17% reporting radiating limb pain, and 2% having been hospitalised due to LBP Trunk extension/flexion was not associated with development of first time LBP	Age, weight and BMI similar between groups Height, occupational physical demands, and occupational musculoskeletal loading was significantly associated with first time LBP ($p<0.05$)

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Table A4. Continued.

Reference	Participants	Testing	Results	Comments
Chaffin [172]	Pre-employed plant workers in a variety of jobs involving manual lifting, $n = 551$	Isometric MVC for torso, arm and leg lift in addition to job specific demands was conducted at baseline Preventative effectiveness of strength relative to job demands were evaluated by examining incidence and severity of low back injuries over an 18 month follow-up period Participants were grouped into tertiles relating to their individual strength relative to their job demands	As job strength requirements exceeded participant strength the incidence and severity of low back injuries increased at a ratio of 3:1 across the tertiles	
Keyserling [173]	Pre-employed plant workers applying for a range of 20 varied jobs, $n = 71$	Isometric MVC for torso and arm lift, and push in/out in addition to job specific demands was conducted at baseline Preventative effectiveness of strength relative to job demands evaluated by placing of experimental ($n = 20$) group into jobs matching strength whereas control group ($n = 51$) were not Incidence of musculoskeletal injuries were evaluated over a 1 year follow-up period	During the follow-up period the control group experienced 19 incidences of musculoskeletal injuries compared to 0 in the experimental group	Age, weight and height similar between groups
Salminen et al. [174]	Healthy children, $n = 38$ Children with LBP, $n = 31$ Children with LBP and sciatica, $n = 7$	Biering-Sorensen test, sit up isometric test with knees at 90^0 and MRI conducted at baseline 3 year follow-up period evaluating LBP ever, LBP in past 12 months, and recurrent/continuous LBP	Both flexion and extension endurance times were significantly lower in LBP groups ($p < 0.05$) at baseline and follow-up yet endurance time was not predictive of development of first time LBP	Age, sex, school matched between groups
Sjolie & Ljunggren [175]	Healthy adolescents, $n = 86$	Biering-Sorensen test and questionnaire regarding LBP conducted at baseline 3 year follow-up period with questionnaire was completed	High mobility/endurance time ratios were significantly associated with development of LBP at follow-up when adjusted for gender, LBP at baseline, and well-being and physical activity at follow-up (OR 1.5–1.9, 95% CI 1.1–3.2, $p < 0.05$)	

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Table A4. Continued.

Reference	Participants	Testing	Results	Comments
Adams et al. [176]	Healthy nurses without history of LBP, $n = 262$ Healthy nurses who had previously suffered with 'non-serious' LBP, $n = 141$	Biering-Sorenson tets, isometric back lift MVC and back lift at 80%MVC for 20 seconds while EMG recorded from T10 and L3 conducted at baseline 3 year follow-up (every 6 months) conducted using questionnaire regarding LBP in previous 6 months	Endurance time at 3 year follow-up was significantly associated with development of serious LBP ($p < 0.01$) and approached significance for any LBP ($p < 0.058$) Neither back lift nor indices of fatigue were associated with development of LBP	
Mostardi et al. [177]	Healthy nurses without history of LBP, $n = 171$	Isokinetic back lift MVC conducted at baseline Injury reports used to examine incidence of low back injury over 2 years follow-up	9% sustained low back injuries during the follow-up period There was no significant difference in strength at baseline between those who reported low back injury during follow-up and those who did not	
Cady et al. [178]	Healthy fire-fighters without LBP, $n = 1652$	Isometric back lift MVC conducted at baseline Incidence of prior low back injuries examined subsequent to baseline measurements – no specific follow-up duration was noted Participants were split into percentiles for 'Most Fit' (84–100 percentile), 'Middle Fit' (17–83 percentile) and 'Least Fit' (0–16 percentile)	7.14% sustained low back injuries in the 'Least Fit' group, 3.19% sustained low back injuries in the 'Middle Fit' group, and 0.77% sustained low back injuries in the 'Most Fit' group	Mean age increased with decreasing fitness levels between the three groups
Mooney et al. [179]	Workers without history of LBP in a ship-building firm in the 3 highest Physical Demand Characteristic categories across 32 jobs, $n = 152$	Isolated lumbar extension MVC using MEDX 2 year follow-up of low back injury and LBP claims	9% sustained low back injuries during the follow-up period the majority occurring in the heavy PDC category (64%) Isolated lumbar extension strength was not predictive of low back injuries and only 2 of those participants injured had below normal strength	Age, height and weight was similar amongst PDC categories and in those injured and uninjured Low back injury rates were significantly higher in heavy and very heavy PDC categories ($p < 0.0001$)
Stevenson et al. [181]	Spinning operators from DuPont without history of LBP, $n = 72$ Spinning operators from DuPont suffering from LBP in previous 2 years, $n = 46$ Spinning operators from DuPont suffering from LBP in previous year, $n = 31$	EMG recorded bilaterally from lumbar paraspinal muscles at T10 and L3 level 3–4 cm from midline during Biering-Sorenson test 2 year follow-up period at 6 month intervals for LBP experiences in previous 6 months	EMG indices of fatigues entered final model and were significantly predictive of LBP ($p = 0.035$)	Other factors in final predictive model included age, peak thoracic acceleration, leg strength/endurance, however psychosocial factors were largely absent.

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Table A4. Continued.

Reference	Participants	Testing	Results	Comments
Heydari et al. [182]	Healthy participants classified as either 'No History of LBP', 'CLBP' or 'Past History of LBP', $n=105$	EMG recorded bilaterally from lumbar paraspinal muscles at L4/L5 level during back lift test maintaining 2/3MVC for 30 seconds at baseline and follow-up 2 year follow-up participants were asked to classify themselves as 'worse', 'better, or 'the same'	At follow-up 76 classified themselves as 'the same', 13 'better' and 16 'worse' EMG indices of fatigue showed greater fatigue was significantly associated with development of first time LBP and with self-classification at follow-up ($p<0.05$)	